Symmetry Collapse in Quantum Computation

Exact Packet Simulation, Polynomial-Time Classical Regimes, and the Boundary of Global Interference

A theorem, benchmark, and research monograph

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Abstract

Quantum circuits are often described in the full 2^n-dimensional Hilbert space even when the input, gates, and measurement occupy a much smaller invariant representation. This monograph develops a precise theory of *symmetry collapse*: the replacement of an apparently exponential quantum evolution by an exact or controlled classical computation on symmetry packets, orbit quotients, occupation coordinates, or another efficiently maintained representation.

The foundational theorem states that a polynomial-size uniform quantum circuit family lies in P whenever an efficiently discoverable invariant packet contains the initial state, has polynomial dimension, admits polynomial-time gate updates, and supports polynomial-time evaluation of the final measurement. Fixed-particle-number circuits at constant particle number provide the basic example. A recorded benchmark at twenty qubits reduces 1,048,576 full amplitudes to 190 exact weight-two amplitudes, with zero reconstruction error in the common arithmetic path and large repeated-application and memory gains.

The framework is then extended. Time-dependent packet schedules and moving frames permit exact simulation even when no single laboratory-frame sector is invariant. Sparse packet-transfer graphs show that a bounded collection of low-excitation sectors remains polynomial despite controlled symmetry breaking. Permutation-symmetric qubit circuits collapse to an $(n+1)$-dimensional Dicke representation, while permutation-invariant mixed-state dynamics admit a stronger Schur-commutant compression: for fixed local dimension d, the complete invariant operator record has size $O_d(n^{d^2-1})$, and for qubits it uses $O(n^3)$ block parameters under efficient access hypotheses. Fixed-local-dimension qudit symmetry has dimension $\binom{n+d-1}{d-1}$. Quantum walks with polynomially many symmetry orbits collapse to quotient dynamics. Few-boson and few-fermion sectors are polynomial for fixed particle number, with a precise phase diagram from polynomial through quasipolynomial and subexponential to exponential growth. Density matrices require two packet labels, channels require four, and approximate moving packets admit cumulative leakage bounds. A second simulation route is also formalized: an exponentially large packet can remain classically tractable when its states admit a compact representation closed under the gate set, as in stabilizer, Gaussian, or bounded-bond tensor-network models.

The limits are equally important. Parity and weight-modulo-four packets provide only constant-factor reductions. Half-filled sectors remain exponential. Stabilizer syndromes do not by themselves compress the logical state. Block encoding does not imply efficient classical access. Forrelation illustrates the opposite regime, where the answer is stored in coherent relations across exponentially many paths rather than in one small invariant sector. The document therefore separates four notions that are often conflated: state-space reduction, practical acceleration, polynomial-time classical simulation, and a universal collapse of quantum complexity. Only the first three are established for declared subclasses. No claim that $P = BQP$ is made.

All figures are generated in Python. The accompanying source contains the figure generator, packet utilities, exact small-instance validation, and archival benchmark data.

Status convention and record basis

The monograph is grounded in the quantum packet and transfer records supplied with the project. Four statuses remain distinct.

Proved statement

Green boxes contain direct proofs, including the invariant-basis theorem, the polynomial packet-simulation theorem, fixed-charge corollaries, and exact operator-block identities.

Derived theorem or corollary

Gold boxes contain deductions made explicit here from the same theorem architecture. The permutation-symmetric, orbit-quotient, few-boson, few-fermion, and compressed-packet results are of this type. Their computational hypotheses are stated rather than hidden.

Recorded exact computation

Blue boxes contain benchmark values or exact finite checks imported from the prior record. They are not reinterpreted as asymptotic theorems. The release code independently verifies the core dimension formulas and exact packet reconstruction on representative instances.

Open program or boundary

Red boxes identify boundaries and research programs, including efficient discovery of hidden packets, hybrid packet–tensor compression, approximate-symmetry compilation, and representation-independent lower bounds on global interference compression.

Scope boundary

The phrase “collapse to classical polynomial time” means membership in the classical complexity class P for the declared circuit subclass and access model. It does not mean equal polynomial exponents, equal wall-clock time, or a proof that every quantum circuit is classically efficient.

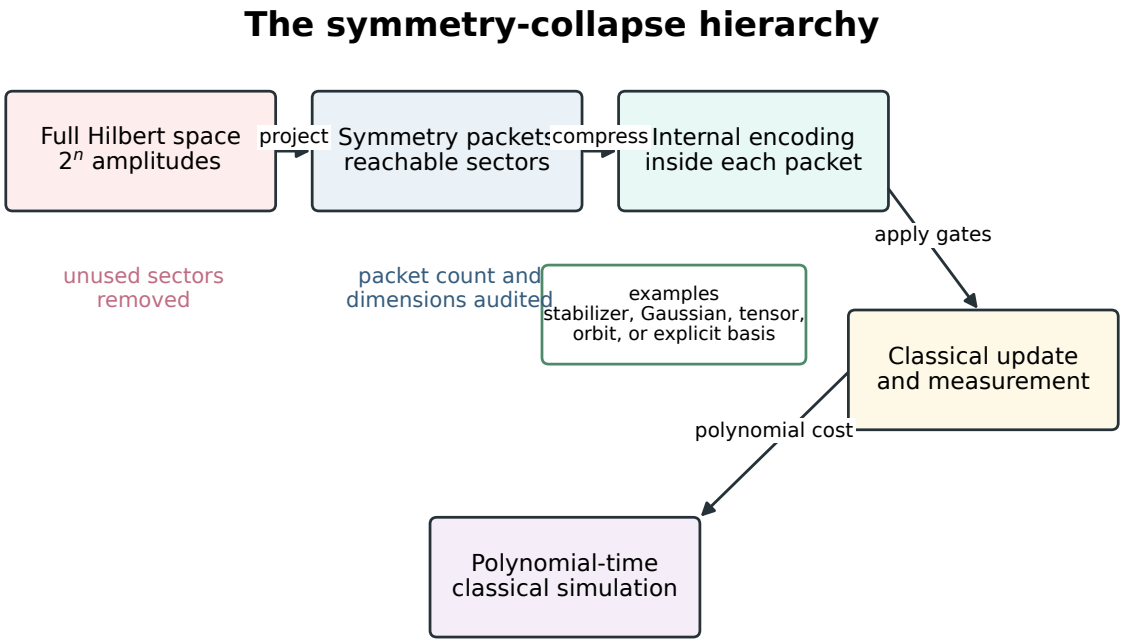


Figure 1: The reduction hierarchy used throughout the monograph.

Executive summary

The records already contain a rigorous classical-collapse theorem. If a uniform family of polynomial-size quantum circuits evolves inside a polynomial-dimensional invariant packet, and the packet basis, restricted gates, and final measurement are all computable in polynomial time, then the accepted language lies in P. The proof is not a physical approximation. It is ordinary deterministic simulation of a polynomial-length vector by polynomial-size matrices.

This theorem has several exact consequences.

1. For fixed k, number-preserving circuits initialized in the weight-k sector are classically polynomial-time simulable because $\binom{n}{k} = O(n^k)$.
2. Exact conservation is stronger than necessary: a circuit that moves among a polynomial-dimensional schedule of sectors, or remains inside one small packet in an efficiently computable moving frame, is also in P.
3. Circuits confined to a fixed union of low-excitation sectors are polynomial even when individual gates create or annihilate particles.
4. Fully permutation-symmetric qubit circuits initialized and measured symmetrically are classically polynomial-time simulable in an $(n+1)$-dimensional Dicke basis, provided the collective gate matrices are efficiently available.
5. More generally, fixed-d permutation-invariant mixed-state dynamics and measurements admit a polynomial Schur-commutant record; the qubit record has $O(n^3)$ density-matrix parameters under efficient block access.
6. Fixed-local-dimension symmetric qudit circuits are polynomial because $\dim \mathrm{Sym}^n(\mathbb{C}^d) = \binom{n+d-1}{d-1}$.
7. Quantum walks reduce to polynomial quotient dynamics when the initial state and measurement are orbit invariant, the number of relevant orbits is polynomial, and quotient couplings are computable without enumerating the full graph.
8. Fixed-number bosonic and fermionic systems are polynomial for a fixed number of particles and polynomially many modes, subject to efficient gate updates. Slowly growing particle number gives quasipolynomial or subexponential, rather than polynomial, explicit-amplitude simulation.

The benchmark record gives a practical demonstration. At $n = 20$ and exact weight two, the full state has 1,048,576 amplitudes while the packet state has 190. The archived repeated-application gain is approximately 958.8 and the amplitude-memory ratio is 5518.8, with zero reported reconstruction error in the common arithmetic path. These values are implementation evidence, not universal machine-independent speedup constants.

The main negative conclusion is just as important. A symmetry label, a moving frame, or a short Schur label does not automatically imply polynomial simulation. Parity leaves 2^{n-1} states. Half filling leaves $\binom{n}{n/2} \sim 2^n/\sqrt{n}$. Entanglement may remain difficult inside one packet. Global interference may

encode the answer in relations between exponentially many contributions. A correct collapse claim must therefore account for packet discovery, internal representation, transfer sparsity, leakage, and measurement.

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Part I

Foundations

Chapter 1

The symmetry-collapse question

1.1 The computational mismatch

An n-qubit pure state has 2^n complex amplitudes in the computational basis. That description is universal but frequently redundant. A number-preserving circuit started with two excitations never visits basis strings of weight zero, one, three, or more. A permutation-symmetric circuit started in a symmetric state never distinguishes two strings that differ only by a relabeling of qubits. A quantum walk on a highly symmetric graph may evolve entirely in the span of a polynomial number of orbit states. In each case the full Hilbert space is a valid container but not the intrinsic state space of the computation.

The question is therefore not whether a quantum circuit acts on an exponential ambient space. It is whether the reachable part of that space, together with the gate action and measurement, has a polynomial classical representation.

Definition 1.1 (Exact symmetry collapse)**.** A uniform quantum circuit family exhibits an exact symmetry collapse when its acceptance probability can be computed by an exact classical simulation on an invariant representation whose description and update costs are polynomial in the input length.

Definition 1.2 (Approximate symmetry collapse)**.** A uniform circuit family exhibits an approximate symmetry collapse to error $\varepsilon(n)$ when a polynomial-size classical representation reproduces its acceptance probability to additive error $\varepsilon(n)$, with the error below the promise gap required by the decision problem.

1.2 Four conclusions that must not be conflated

The phrase "symmetry reduction" is used for several logically different achievements.

1. **Structural reduction.** A projector, quotient, or representation-theoretic coordinate change removes redundant coordinates.
2. **Practical acceleration.** A concrete implementation uses less time or memory on tested instances.
3. **Polynomial classical simulation.** A uniform family admits a deterministic polynomial-time classical algorithm.
4. **Universal complexity collapse.** Every polynomial-size quantum circuit admits such a representation.

The first three occur in this monograph for declared subclasses. The fourth would imply $BQP \subseteq P$ and is not established. A benchmark can support the second conclusion without proving the third. A small invariant label set can support the first conclusion while leaving exponentially large internal blocks and therefore failing to prove the third.

1.3 The four audit quantities

For a proposed reduction, define the schematic complexity record

$$\mathfrak{C}_n = (N_{\text{packets}}, D_{\text{internal}}, S_{\text{transfer}}, L_{\text{leakage}}).$$

Here:

- N_{packets} is the number of reachable symmetry labels.
- D_{internal} is the packet dimension or the size of a compressed internal representation.
- S_{transfer} records the sparsity and cost of gate-induced transitions.
- L_{leakage} measures departure from the proposed representation.

A complete claim must also state the cost of discovering this structure from the input. A decomposition that is small but exponentially hard to find does not yield a polynomial algorithm.

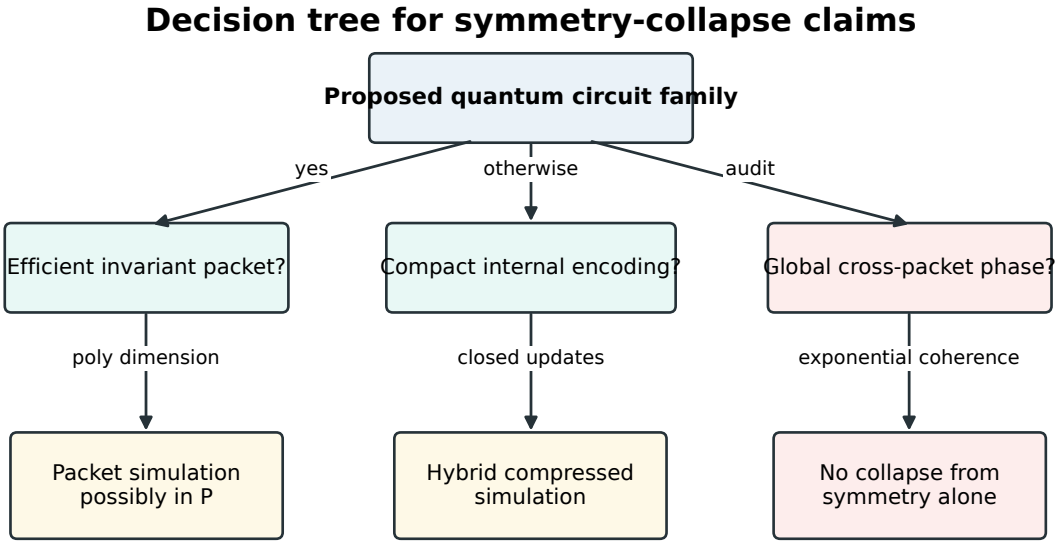


Figure 1.1: A disciplined decision tree for proposed symmetry-collapse claims.

1.4 Why symmetry is only the beginning

Suppose a group G acts on the Hilbert space and the circuit commutes with this action. Representation theory decomposes the space into isotypic pieces. This gives a canonical block structure, but three questions remain.

First, how many pieces are reachable from the input? Second, how large is each piece or how compactly can its states be encoded? Third, can each gate and measurement be applied in those coordinates without reconstructing the full state? The monograph repeatedly returns to this three-stage distinction:

$$\text{symmetry label} \longrightarrow \text{reachable representation} \longrightarrow \text{efficient classical algorithm.}$$

Principle 1.3 (No free collapse)**.** *Block diagonality is a structural statement. Polynomial-time classical simulation requires an efficient basis or encoding, efficient updates, and efficient output evaluation in addition to the block decomposition.*

Chapter 2

Projectors and quantum packets

2.1 Orthogonal packet decompositions

Let $\mathcal{H}$ be a finite-dimensional Hilbert space and let $\{P_\alpha\}_{\alpha\in A}$ be orthogonal projectors satisfying

$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \qquad \sum_{\alpha\in A} P_\alpha = I.$$

Every state and operator decompose as

$$|\psi\rangle = \sum_\alpha P_\alpha |\psi\rangle, \qquad X = \sum_{\alpha,\beta} P_\alpha X P_\beta.$$

The diagonal blocks act within packets. The off-diagonal blocks transfer amplitude or coherence between packets.

Define the packet-transfer graph of an operator X by placing an edge $\beta \to \alpha$ whenever

$$P_\alpha X P_\beta \neq 0.$$

An exactly symmetry-preserving operator has only diagonal edges. A sparse transfer graph may still permit efficient simulation if the active packets and internal states remain compact.

2.2 Conserved observables

Let

$$Q = \sum_k q_k P_k$$

be a self-adjoint observable with spectral projectors P_k.

Theorem 2.1 (Invariant quantum packets)**.** *If a unitary U satisfies $[U, Q] = 0$, then*

$$P_i U P_j = 0 \qquad (i \neq j),$$

and

$$U = \bigoplus_i U_i, \qquad U_i = P_i U P_i\big|_{P_i\mathcal{H}}.$$

If $P_k|\psi\rangle = |\psi\rangle$, then $P_k U|\psi\rangle = U|\psi\rangle$.

Proof. Since U commutes with Q, it commutes with every spectral projector obtained from the functional calculus of Q. Thus

$$P_i U P_j = U P_i P_j = 0$$

for $i \neq j$. Summing the diagonal blocks gives the direct-sum form. The final statement follows from

$$P_k U|\psi\rangle = U P_k|\psi\rangle = U|\psi\rangle.$$

□

2.3 Residue and character packets

Suppose Q has integer spectrum. For a modulus M, define

$$B_r^{(M)} = \sum_{k \equiv r \pmod{M}} P_k.$$

Let

$$V_M = \exp\left(\frac{2\pi i}{M} Q\right), \qquad \omega_M = e^{2\pi i/M}.$$

Then

$$B_r^{(M)} = \frac{1}{M} \sum_{\ell=0}^{M-1} \omega_M^{-r\ell} V_M^{\ell}. \tag{2.1}$$

On a Q-eigenvector, the right side is the roots-of-unity filter. The same finite-character formula selects residue classes in signal polyphase analysis and charge sectors in quantum theory.

2.4 Finite-group projectors

Let a finite group G act unitarily by U_g. For an irreducible character χ of degree d_χ, the central isotypic projector is

$$P_\chi = \frac{d_\chi}{|G|} \sum_{g \in G} \overline{\chi(g)} U_g. \tag{2.2}$$

Character orthogonality gives

$$P_\chi P_\psi = \delta_{\chi\psi} P_\chi, \qquad \sum_\chi P_\chi = I$$

on the represented space. If an operator A commutes with every U_g, then

$$P_\chi A P_\psi = 0 \qquad (\chi \neq \psi).$$

This framework includes momentum, parity, total spin, point-group symmetry, permutation symmetry, and stabilizer syndrome labels. The isotypic projector identifies a symmetry type. It does not necessarily resolve the multiplicity space within that type.

2.5 Nested refinements

If one conserved label is too coarse, intersect commuting projector families. For mutually commuting observables $Q_1, \dots, Q_s$, a joint packet has the form

$$P_{\boldsymbol{\alpha}} = P^{(1)}_{\alpha_1} \cdots P^{(s)}_{\alpha_s}.$$

The dimension can decrease dramatically, but each extra label must be efficiently computable and preserved. A refinement that requires exponential preprocessing merely relocates the complexity.

Remark 2.2. A packet decomposition is basis independent as an orthogonal direct sum, although a practical simulator must choose a concrete basis or compressed coordinate system within each active packet.

Chapter 3

Uniform circuits and the classical cost model

3.1 Circuit families

A quantum decision procedure is a polynomial-time uniform family of circuits $\{C_x\}$, where the input string x determines a circuit on $n = n(|x|)$ qubits with at most $\mathrm{poly}(|x|)$ gates. Let

$$C_x = G_m \cdots G_1.$$

A classical simulation theorem must describe an algorithm receiving the same succinct description of x and the circuit. It is not enough to assume that an exponentially large matrix or a complete orbit list has already been supplied.

3.2 The packet access model

For each input length, suppose there is a packet projector P_n and a basis

$$\mathcal{B}_n = \{|b_1\rangle, \ldots, |b_{D_n}\rangle\}$$

of $P_n\mathcal{H}_n$. The simulation interface has four routines.

1. **Discovery and indexing.** Produce or index $\mathcal{B}_n$ in polynomial time.
2. **Initialization.** Express the initial state in this basis.
3. **Gate update.** Apply each restricted gate $P_nG_jP_n$ to the packet coordinates in polynomial time.
4. **Measurement.** Compute the final acceptance probability to the needed precision.

The basis need not be listed explicitly if there is a compact data structure supporting these operations. Stabilizer tableaux and covariance matrices are examples of representations that encode exponentially many computational-basis amplitudes without storing them individually.

3.3 Precision and promise gaps

A BQP decision problem has an inverse-constant promise gap after standard amplification. A deterministic classical simulator need only estimate the acceptance probability to an additive error

smaller than this gap. If the gate entries are algebraic or otherwise polynomial-time computable, ordinary numerical analysis gives polynomial bit complexity for polynomially many updates, provided condition numbers and error amplification are controlled.

3.4 State-vector cost inside a packet

If the simulator stores D amplitudes and a local gate can be applied in $O(nD)$ arithmetic operations, a depth-d packet simulation has schematic cost

$$C_{\text{packet}} = O(dnD) + C_{\text{compile}}.$$

The corresponding full state-vector path is

$$C_{\text{full}} = O(dn2^n).$$

These formulas are intentionally schematic. Cache behavior, indexing, vectorization, and basis compilation determine actual runtime. The asymptotic distinction is exact when D is polynomial and 2^n is exponential.

3.5 Reuse and amortization

Packet compilation can be expensive. Repeated parameter sweeps, time stepping, variational optimization, and repeated likelihood evaluation benefit from cached basis maps and gate plans. A one-shot simulation can be slower even when each subsequent packet update is much cheaper. The benchmark chapter preserves both one-shot and repeated-application measurements for this reason.

Scope boundary

A polynomial-dimensional invariant sector does not guarantee a practical speedup at small n. It guarantees that the state-vector representation within that sector has polynomial asymptotic size under the stated access assumptions.

Chapter 4

Exact packet simulation and polynomial collapse

4.1 Gatewise invariance

Let P be an orthogonal projector and suppose the initial state obeys $P|\psi_0\rangle = |\psi_0\rangle$. If every gate satisfies

$$G_j P = P G_j P, \tag{4.1}$$

then G_j maps $P\mathcal{H}$ into itself. The equation is slightly more general than commutation: it requires invariance of the selected subspace, not necessarily invariance of its orthogonal complement.

Theorem 4.1 (Invariant-basis simulation)**.** *Let $U = G_m \cdots G_1$ and assume Equation (4.1) for every gate. If $P|\psi_0\rangle = |\psi_0\rangle$, then*

$$U|\psi_0\rangle = (PG_mP)\cdots(PG_1P)|\psi_0\rangle.$$

In any basis of $P\mathcal{H}$, the restricted circuit gives exactly the same amplitudes after embedding.

Proof. Inductively, if $P|\psi_{j-1}\rangle = |\psi_{j-1}\rangle$, then

$$G_j|\psi_{j-1}\rangle = G_jP|\psi_{j-1}\rangle = PG_jP|\psi_{j-1}\rangle,$$

so the next state remains in $P\mathcal{H}$. Iterating proves the identity. □

4.2 Polynomial packet-simulation theorem

Theorem 4.2 (Polynomial packet-simulation theorem)**.** *Let $\{C_x\}$ be a polynomial-time uniform family of quantum circuits. Suppose there is an efficiently computable packet representation such that:*

1. *the initial state lies in the represented packet;*
2. *every gate preserves the representation;*
3. *the representation size is at most* $\mathrm{poly}(|x|)$*;*
4. *each gate update is computable to polynomial precision in polynomial time;*
5. *the final acceptance probability is computable from the representation in polynomial time.*

Then the language decided by the circuit family belongs to P.

Proof. The simulator stores a polynomial-size classical description of the evolving state. It performs polynomially many polynomial-time gate updates and evaluates the final measurement to inverse-polynomial precision. Choosing an error smaller than the amplified promise gap yields a deterministic polynomial-time decision algorithm. □

Proved statement

The theorem is a genuine quantum-to-classical complexity collapse for the declared subclass. It does not assume that a full 2^n state vector is ever constructed.

4.3 Dimension-only corollary

A convenient special case uses an explicit basis of a packet of dimension D_n.

Corollary 4.3 (Polynomial-dimensional invariant packet)**.** *If $D_n \leq \mathrm{poly}(n)$, the packet basis is efficiently indexed, every gate has a polynomial-time sparse or structured action in that basis, and the measurement is efficiently evaluable, then the circuit family is classically polynomial-time simulable.*

4.4 A broader compressed representation

The state description need not be a vector of packet amplitudes. Let $\mathcal{R}_n$ be a polynomial-size classical representation class with an embedding into the packet Hilbert space. If initialization, gate updates, and measurement stay within $\mathcal{R}_n$ and are efficient, the same proof applies. This observation will later combine packet labels with stabilizer, Gaussian, or tensor-network coordinates.

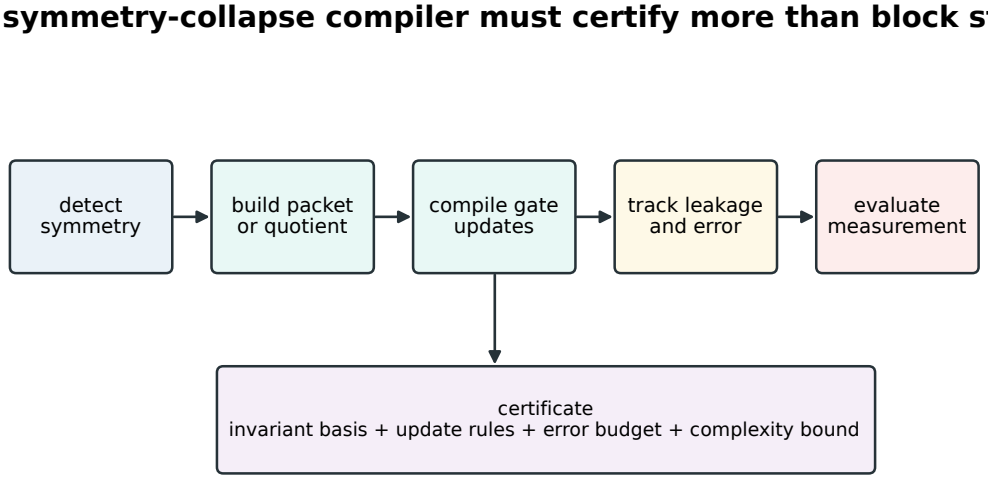


Figure 4.1: A complete collapse certificate includes discovery, update, leakage, measurement, and complexity information.

Chapter 5

Approximate symmetry and leakage certificates

5.1 Hamiltonian leakage

Exact conservation is not the only useful regime. Let P be an orthogonal projector and let

$$U(t) = e^{-itH}$$

for a bounded self-adjoint Hamiltonian H. The leakage operator from the packet is

$$L_P(t) = (I - P)e^{-itH}P.$$

Theorem 5.1 (Commutator leakage bound)**.** *For every real t,*

$$\|(I - P)e^{-itH}P\| \leq |t|\,\|[H, P]\|.$$

Proof. Define

$$F(s) = e^{-i(t-s)H}Pe^{-isH}P.$$

Then $F(0) = e^{-itH}P$ and $F(t) = Pe^{-itH}P$. Differentiation gives

$$F'(s) = ie^{-i(t-s)H}[H, P]e^{-isH}P.$$

Therefore

$$(I - P)e^{-itH}P = -(I - P)\int_0^t F'(s)\,ds.$$

The unitary factors and projectors have norm at most one, so the integral is bounded by $|t|\|[H, P]\|$. □

5.2 Circuit leakage

For a gate G_j, define

$$\delta_j = \|(I - P)G_jP\|.$$

Let $V_j = PG_jP$. A telescoping expansion gives

$$G_m \cdots G_1P - V_m \cdots V_1P = \sum_{k=1}^{m} G_m \cdots G_{k+1}(G_k - V_k)V_{k-1} \cdots V_1P.$$

Taking norms yields the following.

Proposition 5.2 (Sum-of-leakages bound)**.** *For an initial state in $P\mathcal{H}$,*

$$\|G_m \cdots G_1|\psi_0\rangle - V_m \cdots V_1|\psi_0\rangle\| \le \sum_{j=1}^{m} \delta_j.$$

This is conservative because leakage contributions may interfere destructively. Its value is that it is composable and can be checked gate by gate.

5.3 Decision-problem consequence

If the packet representation is polynomial, the restricted updates are efficient, and

$$\sum_j \delta_j \le \frac{1}{\mathrm{poly}(n)},$$

then the approximate packet simulator reproduces the final acceptance probability to inverse-polynomial accuracy. After ordinary promise-gap amplification, this suffices for a deterministic polynomial-time decision algorithm.

5.4 Adaptive packet enlargement

A practical simulator need not fix one packet forever. It can maintain an active set A_t of packets, estimate transfers $P_\alpha G_t P_\beta$, and add a packet when its predicted mass exceeds a tolerance. The proof obligation becomes a global accounting identity:

$$\text{total error} \le \text{discarded transfer} + \text{numerical error} + \text{measurement approximation}.$$

The key research problem is to retain polynomial active size while certifying the tail.

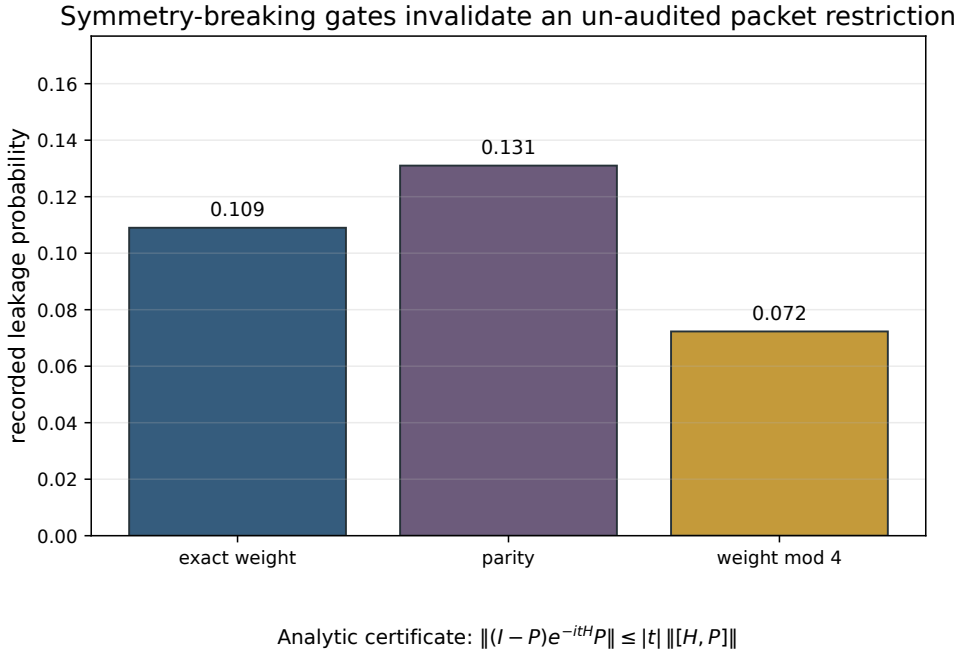


Figure 5.1: Recorded leakage examples and the analytic commutator certificate.

Chapter 6

Moving packets, dynamic frames, and sparse transfer graphs

6.1 Why one fixed projector is not the end of the theory

The exact packet theorem in Chapter 4 follows a state inside one fixed subspace. That is sufficient for a conserved charge, but it is not necessary for classical collapse. A circuit may move through a sequence of small subspaces, or it may preserve a small packet only after a time-dependent change of coordinates. The correct object is therefore a *packet schedule*, not necessarily one static projector.

Definition 6.1 (Packet schedule)**.** For a circuit

$$U = G_m \cdots G_1$$

on a Hilbert space $\mathcal{H}$, a packet schedule is a sequence of orthogonal projectors

$$P_0, P_1, \dots, P_m.$$

It is exact when

$$G_t P_{t-1} = P_t G_t P_{t-1} \qquad (1 \le t \le m). \tag{6.1}$$

The condition says that gate G_t maps the currently active packet into the next active packet. The dimensions may vary with time.

6.2 Exact dynamic-packet simulation

Choose isometries

$$B_t : \mathbb{C}^{D_t} \longrightarrow \mathcal{H}, \qquad B_t B_t^* = P_t, \qquad B_t^* B_t = I_{D_t},$$

where $D_t = \operatorname{rank} P_t$. Define the rectangular coordinate update

$$M_t = B_t^* G_t B_{t-1}. \tag{6.2}$$

Theorem 6.2 (Exact dynamic-packet theorem)**.** *Suppose $P_0|\psi_0\rangle = |\psi_0\rangle$ and Equation (6.1) holds for every gate. Put*

$$z_0 = B_0^*|\psi_0\rangle, \qquad z_t = M_t z_{t-1}.$$

Then

$$G_t \cdots G_1|\psi_0\rangle = B_t z_t$$

for every t.

Proof. The claim is true at $t = 0$. If the state after $t - 1$ gates is $B_{t-1}z_{t-1}$, then

$$G_t B_{t-1} z_{t-1} = P_t G_t P_{t-1} B_{t-1} z_{t-1} = B_t B_t^* G_t B_{t-1} z_{t-1} = B_t M_t z_{t-1}.$$

This is $B_t z_t$. □

Theorem 6.3 (Polynomial moving-packet collapse)**.** *Let $\{C_x\}$ be a polynomial-time uniform circuit family. Suppose an exact packet schedule can be produced from x such that:*

1. $\max_t D_t \leq \mathrm{poly}(|x|)$*;*
2. *the coordinate description of the initial state is computable in polynomial time;*
3. *each rectangular update M_t can be applied to polynomial precision in polynomial time;*
4. *the final acceptance probability is computable from z_m in polynomial time.*

Then the language decided by the circuit family belongs to P.

Proof. The simulator stores at most polynomially many coordinates at every step and applies polynomially many polynomial-time updates. Theorem 6.2 proves exactness before numerical rounding. Polynomial precision below the amplified promise gap gives a deterministic polynomial-time decision procedure. □

Derived theorem or corollary

A circuit does not need one globally conserved sector. It is enough that the state follow an efficiently discoverable path through polynomial-dimensional sectors.

6.3 Moving frames and hidden symmetry

The packet schedule has a useful coordinate form. Let P be one fixed projector and let W_t be unitaries with

$$P_t = W_t P W_t^*.$$

In the moving frame, the transformed gate is

$$\widetilde{G}_t = W_t^* G_t W_{t-1}. \tag{6.3}$$

The schedule condition is equivalent to

$$\widetilde{G}_t P = P \widetilde{G}_t P.$$

Corollary 6.4 (Moving-frame collapse)**.** *Suppose a polynomial-time uniform circuit admits efficiently described frames W_t for which the transformed gates in Equation (6.3) preserve one fixed polynomial-dimensional packet, and their restricted matrices and the transformed final measurement are efficiently computable. Then the circuit family is classically simulable in polynomial time.*

This is the appropriate theorem for a symmetry that changes orientation with time. Examples to test include Clifford frames, Gaussian mode frames, rotating spin axes, and gauge-fixed local coordinates.

Scope boundary

The frame must be independently discoverable and efficiently usable. Choosing $W_t = G_t \cdots G_1$ makes the transformed state trivial but is circular: computing that frame is the original simulation problem. The theorem is useful only when the frames have a compact classical description and their restricted actions can be computed without constructing the full state.

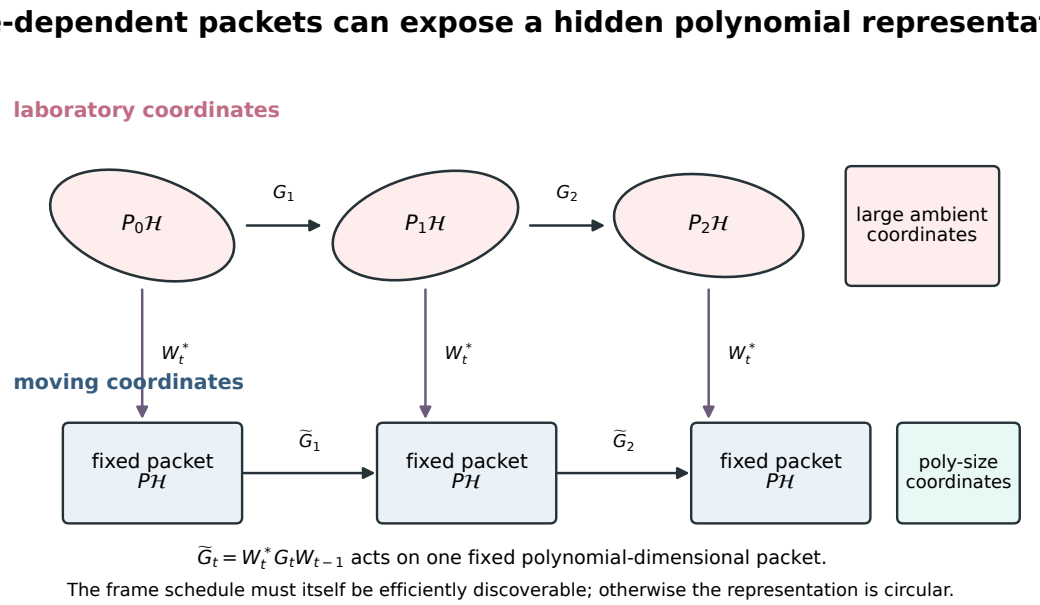


Figure 6.1: A circuit may fail to preserve one laboratory-frame packet while remaining inside a small packet in a computable moving frame.

6.4 Sparse packet-transfer graphs

Let

$$I = \sum_{\alpha \in \Lambda} P_\alpha$$

be an orthogonal packet decomposition. Write the gate blocks as

$$G_{t;\beta\alpha} = P_\beta G_t P_\alpha.$$

Starting from an active packet set S_0, define recursively

$$S_t = \{\beta : \exists \alpha \in S_{t-1} \text{ with } G_{t;\beta\alpha} \neq 0\}. \tag{6.4}$$

Corollary 6.5 (Reachable packet-graph collapse)**.** *Suppose the packet sets S_t and nonzero transfer blocks are computable in polynomial time, and*

$$D_t = \sum_{\alpha \in S_t} \operatorname{rank} P_\alpha \leq \operatorname{poly}(n)$$

for every t. If all active block updates and the final measurement are efficiently computable, then the circuit family is classically polynomial-time simulable.

Proof. Take

$$P_t = \sum_{\alpha \in S_t} P_\alpha.$$

The definition of S_t gives $G_t P_{t-1} = P_t G_t P_{t-1}$. Apply Theorem 6.3. □

This theorem permits exact sector changes. What matters is not that one charge be fixed, but that the union of reachable sectors remain polynomial.

Corollary 6.6 (Bounded-excitation collapse)**.** *Consider n qubits and suppose the initial state and every circuit prefix are certified to lie in*

$$\mathcal{H}_{\le K} = \bigoplus_{k=0}^{K} \mathcal{H}_{n,k}$$

for a fixed constant K. If the gates and measurement have polynomial-time actions in the occupation basis, then the circuit family lies in P, even when individual gates change excitation number.

Proof. The active dimension is

$$\sum_{k=0}^{K} \binom{n}{k} = O(n^K).$$

Use the weight sectors as packet nodes and apply Theorem 6.5. □

For example, a fixed-weight input followed by only a constant number of pair-creation or pair-annihilation events remains in a constant collection of low-weight sectors. Exact number conservation is therefore stronger than necessary.

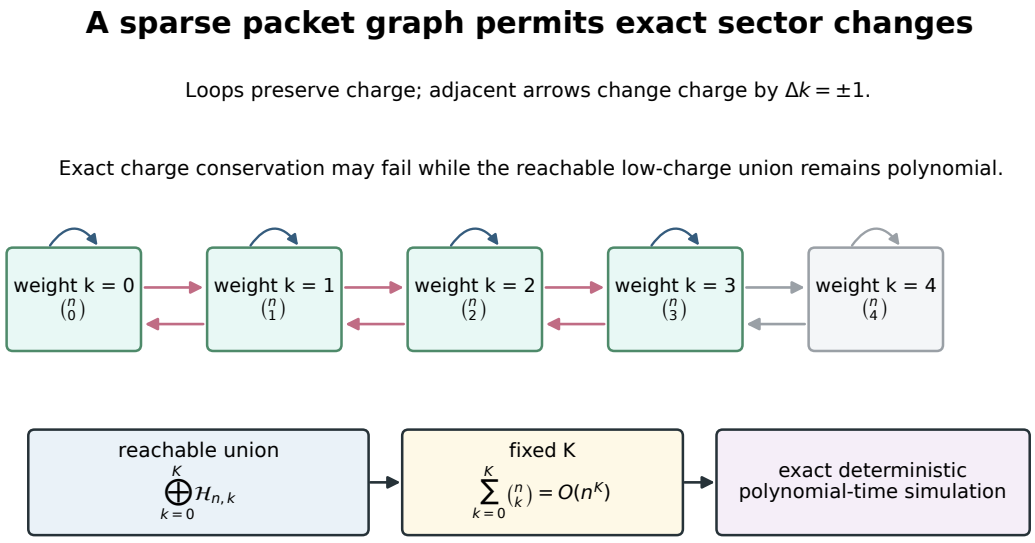


Figure 6.2: Sector-changing gates remain classically tractable when the reachable packet graph and total active dimension stay polynomial.

6.5 Approximate dynamic packets

Define the one-step leakage from the scheduled packet by

$$\delta_t = \|(I - P_t)G_tP_{t-1}\|.$$

Let

$$\widehat{G}_t = P_tG_tP_{t-1}.$$

Theorem 6.7 (Dynamic sum-of-leakages bound)**.** *For an initial unit vector in $P_0\mathcal{H}$,*

$$\left\|G_m \cdots G_1|\psi_0\rangle - \widehat{G}_m \cdots \widehat{G}_1|\psi_0\rangle\right\| \leq \sum_{t=1}^{m} \delta_t.$$

For any acceptance operator $0 \leq M \leq I$, the corresponding acceptance probabilities differ by at most

$$2\sum_{t=1}^{m} \delta_t.$$

Proof. A telescoping expansion inserts $G_t - \widehat{G}_t$ at one time. On $P_{t-1}\mathcal{H}$ this difference equals $(I - P_t)G_tP_{t-1}$, while all other unitary or projected factors have norm at most one. Summing the operator norms proves the state bound. If ψ and ϕ have norm at most one, then

$$|\langle\psi, M\psi\rangle - \langle\phi, M\phi\rangle| \leq (\|\psi\| + \|\phi\|)\|\psi - \phi\| \leq 2\|\psi - \phi\|.$$

□

Corollary 6.8 (Approximate moving-packet collapse)**.** *If the scheduled dimensions and updates are polynomial and*

$$\sum_t \delta_t \leq \frac{1}{\mathrm{poly}(n)},$$

then, after ordinary promise-gap amplification, the circuit family remains deterministically classically simulable in polynomial time.

The result is conservative but fully composable. It extends the fixed-projector estimate of Chapter 5 to moving frames and changing packet sets.

Part II

Exact symmetry families

Chapter 7

Fixed charge and exact-number circuits

7.1 Excitation-number packets

For n qubits, define

$$\widehat{N} = \sum_{q=1}^{n} \frac{I - Z_q}{2}.$$

Its eigenvalues are $k = 0, 1, \ldots, n$. The projector onto Hamming weight k is

$$P_{n,k} = \sum_{|x|=k} |x\rangle\langle x|,$$

and

$$\dim P_{n,k}\mathcal{H} = \binom{n}{k}.$$

The decomposition

$$(\mathbb{C}^2)^{\otimes n} = \bigoplus_{k=0}^{n} \mathcal{H}_{n,k}$$

recovers the full dimension through

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

7.2 Number-preserving gates

Single-qubit R_Z gates commute with $\widehat{N}$. A two-qubit exchange rotation acts in the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ as

$$U_{XY}(\theta, \phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -ie^{-i\phi}\sin\theta & 0 \\ 0 & -ie^{i\phi}\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

It rotates $|01\rangle$ and $|10\rangle$ while fixing $|00\rangle$ and $|11\rangle$, so it preserves exact Hamming weight.

Corollary 7.1 (Fixed-charge polynomial simulation)**.** *For every fixed k, a polynomial-time uniform family of number-preserving circuits initialized in $\mathcal{H}_{n,k}$ and measured by an efficiently computable observable is classically simulable in polynomial time, provided each restricted local gate is efficiently applied in the weight-k basis.*

Proof. For fixed k,

$$\dim \mathcal{H}_{n,k} = \binom{n}{k} = O(n^k).$$

The weight-k basis is efficiently indexed by k-element subsets. A local number-preserving gate changes only basis states containing the affected occupation pattern. Apply Theorem 4.2. □

7.3 Sparse update structure

A one-qubit diagonal phase changes one amplitude at a time. An exchange gate on modes p, q couples a basis state only to the state obtained by moving one excitation between those modes. Thus every column of the restricted gate has at most two nonzero entries. For fixed k, the total update cost is polynomial in n.

7.4 Uniformity and input encoding

The fixed-charge corollary assumes k is a fixed constant independent of n. It also applies when $k = O(1)$ is supplied by the circuit family. If k is written in binary and allowed to scale as $n/2$, the sector can become exponential. The theorem is about the dimension and update cost of the actual active sector, not the name of the symmetry.

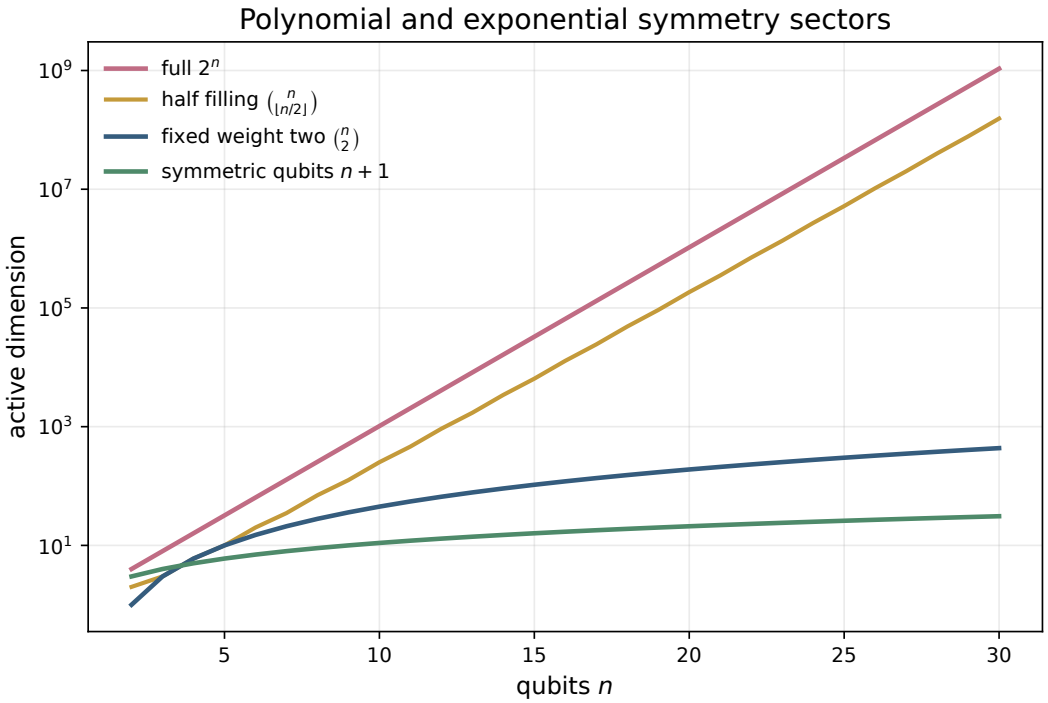


Figure 7.1: Fixed weight and permutation symmetry can be polynomial, while half filling remains exponential.

7.5 Observables

A diagonal observable depending on occupations is immediate. Few-body number-preserving observables are sparse in the subset basis. More global observables may require additional structure. The measurement hypothesis in Theorem 4.2 cannot be omitted: a compactly evolved state does not help if the requested output itself hides exponential work.

Chapter 8

The exact weight-two benchmark

8.1 Purpose and status

The benchmark record compares a full state-vector simulator with a simulator restricted to the exact weight-two packet. Both use the same local gate arithmetic. The benchmark is not a hardware claim and not a universal performance model. It shows that the exact dimension reduction can translate into large memory and repeated-application gains in an ordinary implementation.

Recorded exact computation

At $n = 20$, the full state has 1,048,576 amplitudes and the exact weight-two packet has 190. The archived repeated-application gain is 958.8, the one-shot gain is 54.8, and the amplitude-memory ratio is 5518.8. The recorded relative reconstruction error is zero at the displayed precision.

8.2 Recorded table

Table 8.1: Recorded exact-weight-two benchmark.

n	full dim.	packet dim.	apply gain	one-shot gain	memory gain	rel. error
8	256	28	1.11	1.64	9.14	0
10	1,024	45	1.51	2.02	22.8	0
12	4,096	66	2.95	3.98	62.1	0
14	16,384	91	8.55	11.0	180.0	0
16	65,536	120	33.0	36.9	546.1	0
18	262,144	153	131.3	39.1	1,713.4	0
20	1,048,576	190	958.8	54.8	5,518.8	0

8.3 What is machine independent

The dimension identity

$$\frac{2^n}{\binom{n}{2}}$$

is independent of implementation. It predicts an exponential memory advantage for fixed $k = 2$. The measured runtime ratio need not equal the dimension ratio. Packet indexing, compilation, sparse

update kernels, allocation, and cache behavior determine the constant factors.

8.4 Repeated application versus one shot

Let C_{compile} be the cost of constructing the packet basis and restricted gate plans, and let C_{apply} be one packet application. Reusing the compiled plan R times costs

$$C_{\text{packet}}(R) = C_{\text{compile}} + RC_{\text{apply}}.$$

If the full path costs RC_{full}, the packet method breaks even when

$$R > \frac{C_{\text{compile}}}{C_{\text{full}} - C_{\text{apply}}}.$$

This is why time evolution, parameter sweeps, variational loops, and repeated likelihood evaluations are natural applications.

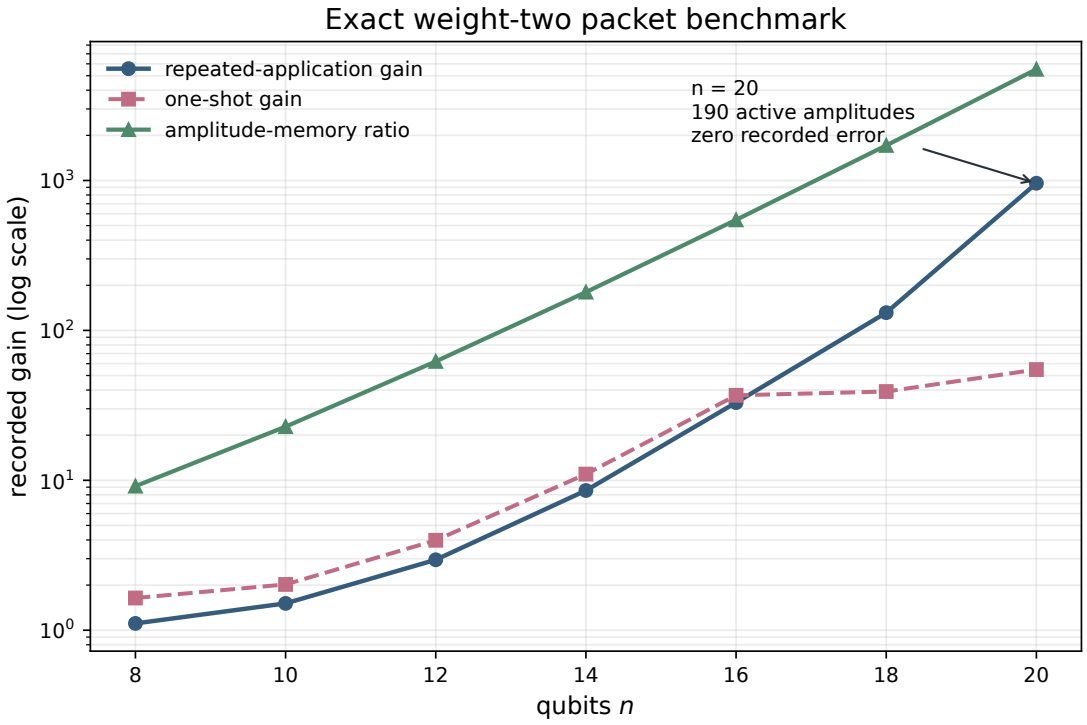


Figure 8.1: Recorded runtime and memory gains for the exact-weight-two packet.

8.5 Independent reconstruction check

The release code generates a random packet state, embeds it in the full Hilbert space, applies the same R_Z and exchange gates through both arithmetic paths, and compares the final states. The representative checks at $(n, k) = (6, 1), (8, 2), (10, 2)$ produce errors near machine precision.

Scope. The recorded benchmark supports exactness and implementation value for a symmetry-preserving subclass. It does not show that generic quantum circuits admit comparable reductions.

Chapter 9

Coarse packets and the honest limitation

9.1 Parity

The even- and odd-parity packets each have dimension

$$2^{n-1}.$$

Parity removes half the state vector. This can produce a useful constant factor, but it does not change exponential scaling.

A pair-creation rotation mixes $|00\rangle$ and $|11\rangle$ and therefore changes excitation number by two while preserving parity. The packet is exact for a parity-preserving circuit, but its internal dimension remains exponential.

9.2 Weight modulo four

A four-qubit rotation between $|0000\rangle$ and $|1111\rangle$ changes excitation number by four. It can mix exact-weight sectors while preserving weight modulo four. A single residue packet contains asymptotically one quarter of all basis states. Again, the reduction is a constant factor.

9.3 Half filling

At half filling,

$$\binom{n}{\lfloor n/2 \rfloor} \sim \frac{2^n}{\sqrt{\pi n/2}}.$$

The packet is smaller by a polynomial factor, but it remains exponential. At $n = 20$, the half-filled packet contains 184,756 amplitudes. The archived repeated-application gain is approximately 7.396, while the one-shot gain is 0.6529, meaning the packet path was slower in that single-run test after compilation overhead.

Recorded exact computation

The half-filled benchmark demonstrates a necessary distinction: dimension reduction can be exact and useful without yielding polynomial complexity or a one-shot speedup.

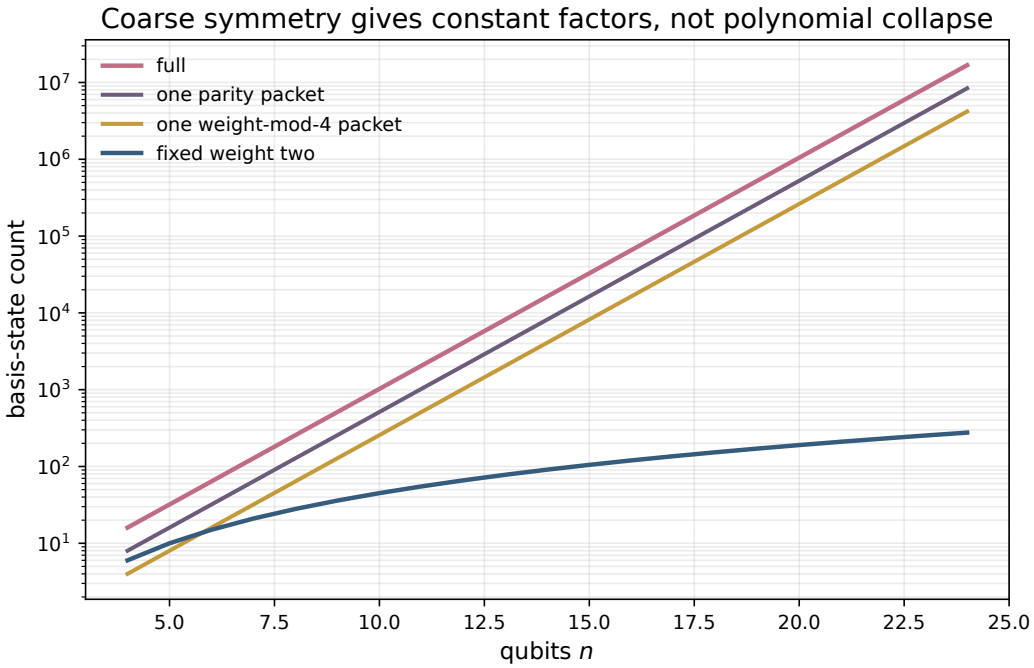


Figure 9.1: Parity and mod-four packets retain an exponential number of basis states.

9.4 Entanglement inside a packet

A small number of symmetry labels does not imply low entanglement. The half-filled packet can support volume-law entanglement. Tensor-network bond dimension and packet dimension are independent resources. A robust simulator should therefore track both

$$(D_{\text{packet}}, \chi_{\text{internal}}),$$

where χ_{internal} measures the cost of representing the state inside the packet.

9.5 The no-label shortcut

A stabilizer syndrome is only a label. A momentum quantum number is only a label. A total charge residue is only a label. The internal multiplicity space may still contain exponentially many degrees of freedom. The packet-simulation theorem becomes polynomial only when the complete active representation, not merely the label set, has polynomial classical cost.

Chapter 10

Permutation-symmetric circuits and Dicke coordinates

10.1 The symmetric subspace

Let the symmetric group S_n act by permuting the n tensor factors. The fully symmetric qubit subspace is spanned by the Dicke states

$$|D_n^k\rangle = \binom{n}{k}^{-1/2} \sum_{|x|=k} |x\rangle, \qquad 0 \le k \le n.$$

Its dimension is $n+1$.

Derived theorem or corollary

If the initial state is permutation symmetric, every gate preserves the fully symmetric subspace, the gate matrices in the Dicke basis are computable in polynomial time, and the final measurement is permutation invariant and efficient, then the circuit family lies in P.

Theorem 10.1 (Permutation-symmetric collapse)**.** *Under the hypotheses above, an n-qubit circuit can be simulated using an $(n+1)$-component classical state vector and polynomial-time updates.*

Proof. The projector onto the trivial S_n representation has rank $n+1$. The Dicke basis is explicitly indexed by $k = 0, \dots, n$. Apply Theorem 4.2. □

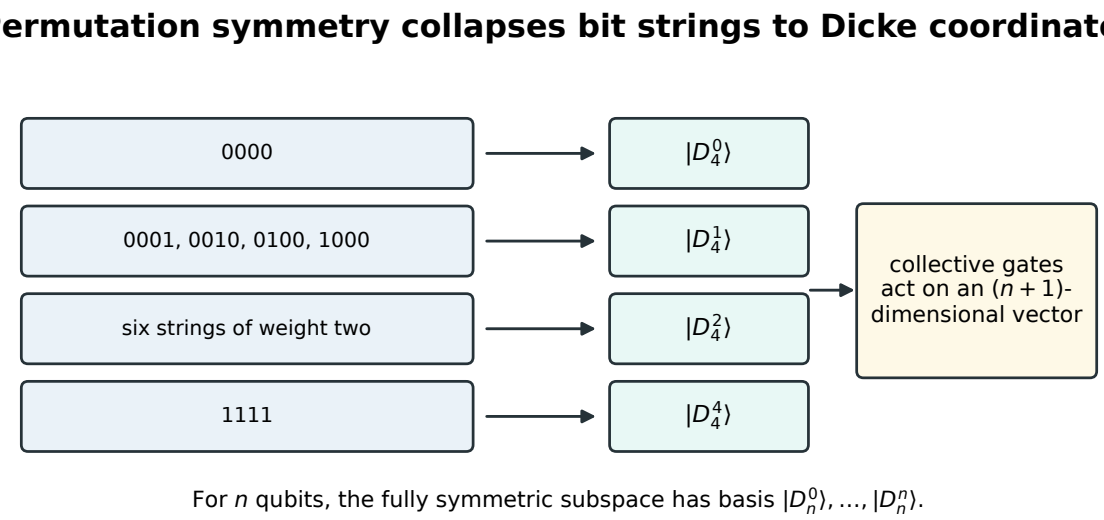


Figure 10.1: Computational-basis orbits become Dicke coordinates.

10.2 Collective spin operators

Define

$$J_a = \frac{1}{2}\sum_{q=1}^{n} \sigma_a^{(q)}, \qquad a \in \{x, y, z\}.$$

The operators J_a preserve the symmetric subspace. In the Dicke basis, J_z is diagonal and

$$J_+|D_n^k\rangle = \sqrt{(n-k)(k+1)}|D_n^{k+1}\rangle,$$

$$J_-|D_n^k\rangle = \sqrt{k(n-k+1)}|D_n^{k-1}\rangle.$$

Thus collective rotations and polynomial functions of collective spin admit banded or structured updates on $n+1$ amplitudes.

10.3 Symmetric qudits

For local dimension d, the symmetric subspace has occupation-number basis

$$(n_1, \ldots, n_d), \qquad n_1 + \cdots + n_d = n,$$

and dimension

$$\dim \mathrm{Sym}^n(\mathbb{C}^d) = \binom{n+d-1}{d-1}. \tag{10.1}$$

For fixed d, this is polynomial in n.

Corollary 10.2 (Fixed-local-dimension symmetric qudits)**.** *A uniform family of fully symmetric d-level circuits is classically polynomial-time simulable for fixed d, provided its occupation-basis updates and measurement are polynomial-time computable.*

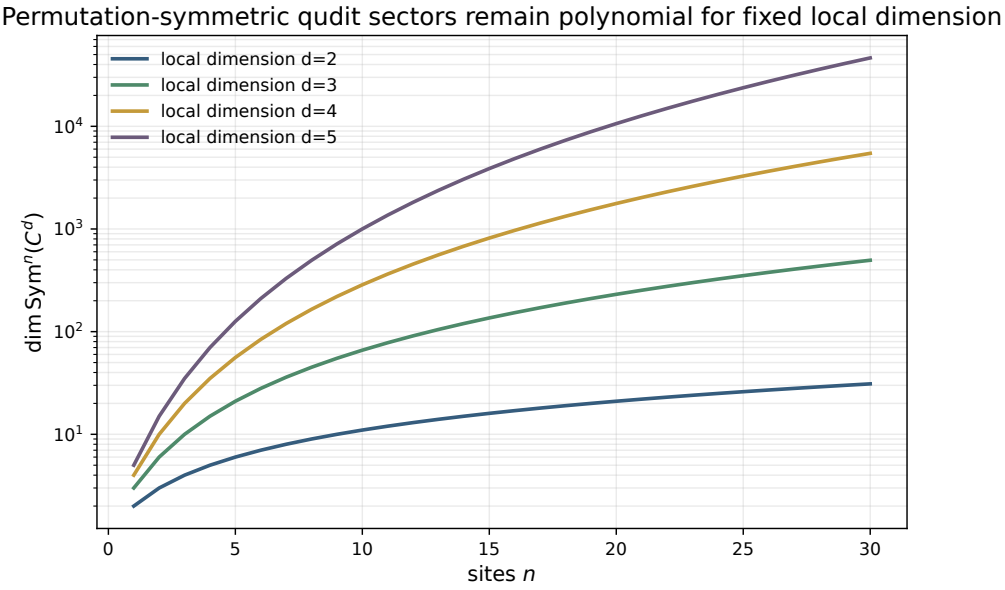


Figure 10.2: Symmetric qudit sectors remain polynomial for every fixed local dimension.

10.4 What full symmetry excludes

A generic two-qubit gate acting on one selected pair does not preserve the fully symmetric subspace. Symmetry is preserved by collective gates or by circuits arranged so that the total operation commutes with all permutations. A gatewise packet simulator requires gatewise invariance, or else it must combine several elementary gates into a symmetry-preserving layer and audit intermediate leakage.

Chapter 11

Schur–Weyl coordinates and total-spin packets

11.1 Decomposition

For n qudits of local dimension d, Schur–Weyl duality decomposes

$$(\mathbb{C}^d)^{\otimes n} \cong \bigoplus_{\lambda \vdash n,\, \ell(\lambda) \le d} \mathcal{U}_\lambda \otimes \mathcal{V}_\lambda,$$

where $\mathcal{U}_\lambda$ carries an irreducible unitary-group representation and $\mathcal{V}_\lambda$ carries a symmetric-group representation. A permutation-invariant operator acts as

$$A = \bigoplus_{\lambda} A_\lambda \otimes I_{\mathcal{V}_\lambda}.$$

The label λ can be polynomially short even when the multiplicity space $\mathcal{V}_\lambda$ is exponentially large. This is a canonical example of the difference between a small packet label and a small active representation.

11.2 Qubit total spin

For qubits, the decomposition may be labeled by total spin j and a multiplicity index. A collective $SU(2)$ operation acts nontrivially only on the $(2j+1)$-dimensional spin factor and as the identity on multiplicity. If the initial state and observable are insensitive to multiplicity, the effective dimension can be small. If arbitrary information is stored in multiplicity, total-spin labeling alone does not compress it.

Principle 11.1 (Multiplicity audit)**.** *A representation-theoretic packet must be recorded as*

$$\textit{irrep label} \quad + \textit{irrep carrier} \quad + \textit{multiplicity space}.$$

Dropping the multiplicity space is valid only when the dynamics, input, and measurement make it redundant.

11.3 Efficient Schur simulation regime

A Schur-coordinate simulation is polynomial when the number of active partitions is polynomial, the retained carrier and multiplicity dimensions are polynomial or compactly represented, and each gate

has an efficient action in these coordinates. Fully symmetric dynamics is the one-row partition and therefore the simplest case.

11.4 Possible compiler use

A symmetry compiler can detect collective layers, transform them into Schur coordinates, apply small representation matrices, and transform only when required. The transformation itself must be costed. In a purely classical simulator, one may never materialize the full Schur transform if the initial state is already specified by occupation or spin coordinates.

Open program or boundary

A broad Schur-based collapse theorem requires explicit uniform algorithms for active-partition discovery, multiplicity handling, and local-gate updates. The decomposition alone is not the algorithm.

Chapter 12

Permutation-invariant mixed states and commutant compression

12.1 Why multiplicity can disappear without being discarded

The fully symmetric subspace is only one Schur–Weyl packet. A general permutation-invariant mixed state can occupy every Schur label while still admitting a polynomial classical description for fixed local dimension. The reason is that the exponentially large symmetric-group multiplicities occur as identity factors rather than independent dynamical coordinates.

Let S_n act on

$$\mathcal{H}_{n,d} = (\mathbb{C}^d)^{\otimes n}$$

by permuting tensor factors. Schur–Weyl duality gives

$$\mathcal{H}_{n,d} \cong \bigoplus_{\lambda \vdash n,\, \ell(\lambda) \le d} \mathcal{U}_\lambda \otimes \mathcal{V}_\lambda, \tag{12.1}$$

where S_n acts trivially on the first tensor factor and irreducibly on $\mathcal{V}_\lambda$.

12.2 The permutation twirl

Define the twirling channel

$$\mathcal{T}_n(X) = \frac{1}{n!} \sum_{\pi \in S_n} U_\pi X U_\pi^*. \tag{12.2}$$

Its image is the commutant of the permutation action.

Lemma 12.1 (Invariant observables see only the twirl)**.** *If A commutes with every U_π, then*

$$\mathrm{Tr}(A\rho) = \mathrm{Tr}\big(A\mathcal{T}_n(\rho)\big)$$

for every density operator ρ.

Proof. Using cyclicity of the trace and $U_\pi^* A U_\pi = A$,

$$\mathrm{Tr}(A\mathcal{T}_n(\rho)) = \frac{1}{n!} \sum_\pi \mathrm{Tr}(AU_\pi \rho U_\pi^*) = \frac{1}{n!} \sum_\pi \mathrm{Tr}(U_\pi^* A U_\pi \rho) = \mathrm{Tr}(A\rho).$$

□

If a unitary U and acceptance operator M are permutation invariant, then U^*MU is invariant. Therefore the final probability depends on the input only through its twirl. This permits an observable-specific collapse even when the original pure input is not itself symmetric.

12.3 Exact commutant record

Let

$$u_\lambda = \dim \mathcal{U}_\lambda, \qquad f_\lambda = \dim \mathcal{V}_\lambda.$$

Schur's lemma gives the following normal form.

Lemma 12.2 (Permutation-invariant state normal form)**.** *A density operator ρ commutes with every permutation if and only if it can be written uniquely as*

$$\rho = \bigoplus_\lambda p_\lambda \, \rho_\lambda \otimes \frac{I_{\mathcal{V}_\lambda}}{f_\lambda}, \tag{12.3}$$

where $p_\lambda \geq 0$, $\sum_\lambda p_\lambda = 1$, and each ρ_λ is a density operator on $\mathcal{U}_\lambda$ whenever $p_\lambda > 0$.

The complete state record is therefore

$$\{(p_\lambda, \rho_\lambda)\}_\lambda .$$

The multiplicity dimension f_λ affects normalization and degeneracy, but no $f_\lambda \times f_\lambda$ matrix needs to be stored.

A permutation-invariant unitary has the form

$$U = \bigoplus_\lambda U_\lambda \otimes I_{\mathcal{V}_\lambda}, \tag{12.4}$$

and a permutation-invariant measurement operator has the analogous form

$$M = \bigoplus_\lambda M_\lambda \otimes I_{\mathcal{V}_\lambda}.$$

The output probability is

$$\operatorname{Tr}(MU\rho U^*) = \sum_\lambda p_\lambda \operatorname{Tr}\big(M_\lambda U_\lambda \rho_\lambda U_\lambda^*\big). \tag{12.5}$$

12.4 Polynomial parameter bound

The number of partitions of n with at most d parts is bounded by the number of weak compositions:

$$\#\{\lambda : \ell(\lambda) \leq d\} \leq \binom{n+d-1}{d-1} = O_d(n^{d-1}).$$

The Weyl dimension formula gives

$$u_\lambda = \prod_{1 \leq i < j \leq d} \frac{\lambda_i - \lambda_j + j - i}{j - i} = O_d\left(n^{d(d-1)/2}\right). \tag{12.6}$$

Therefore

$$\sum_\lambda u_\lambda^2 = O_d\left(n^{d^2-1}\right). \tag{12.7}$$

Theorem 12.3 (Schur-commutant collapse)**.** *Fix the local dimension d. Consider a polynomial-time uniform family of permutation-invariant quantum circuits or channels on n qudits. Suppose:*

1. *the Schur-block record of the twirled input can be constructed in polynomial time;*

2. *the induced update on the invariant operator record in Equation (12.3) is computable to polynomial precision in polynomial time;*
3. *the final measurement is permutation invariant and its Schur blocks are efficiently computable.*

Then the associated decision problem belongs to P.

Proof. Permutation covariance keeps the density operator inside the commutant algebra. By Equation (12.7), that algebra has polynomial dimension for fixed d. The hypotheses supply polynomial-time initialization, updates, and measurement on this record. Apply Theorem 22.1. □

Derived theorem or corollary

For fixed local dimension, permutation-invariant mixed-state dynamics can be classically polynomial even when the state has support across exponentially large symmetric-group multiplicity spaces. Those multiplicities are exact identity factors in the invariant operator record.

12.5 Qubit corollary

For qubits, Schur labels are total spins j. The carrier dimension is

$$u_j = 2j + 1 \leq n + 1,$$

and there are at most $\lfloor n/2 \rfloor + 1$ labels. Hence

$$\sum_j (2j+1)^2 = O(n^3). \tag{12.8}$$

Corollary 12.4 (Permutation-invariant qubit channels)**.** *A uniform permutation-covariant qubit channel family with efficiently computable total-spin block updates, efficiently available twirled input blocks, and a permutation-invariant measurement is classically simulable using $O(n^3)$ density-matrix parameters, up to constant factors and numerical precision.*

Collective Hamiltonians that are polynomial expressions in J_x, J_y, J_z are a basic test class because their total-spin matrices can be generated directly without constructing the full $2^n \times 2^n$ operator.

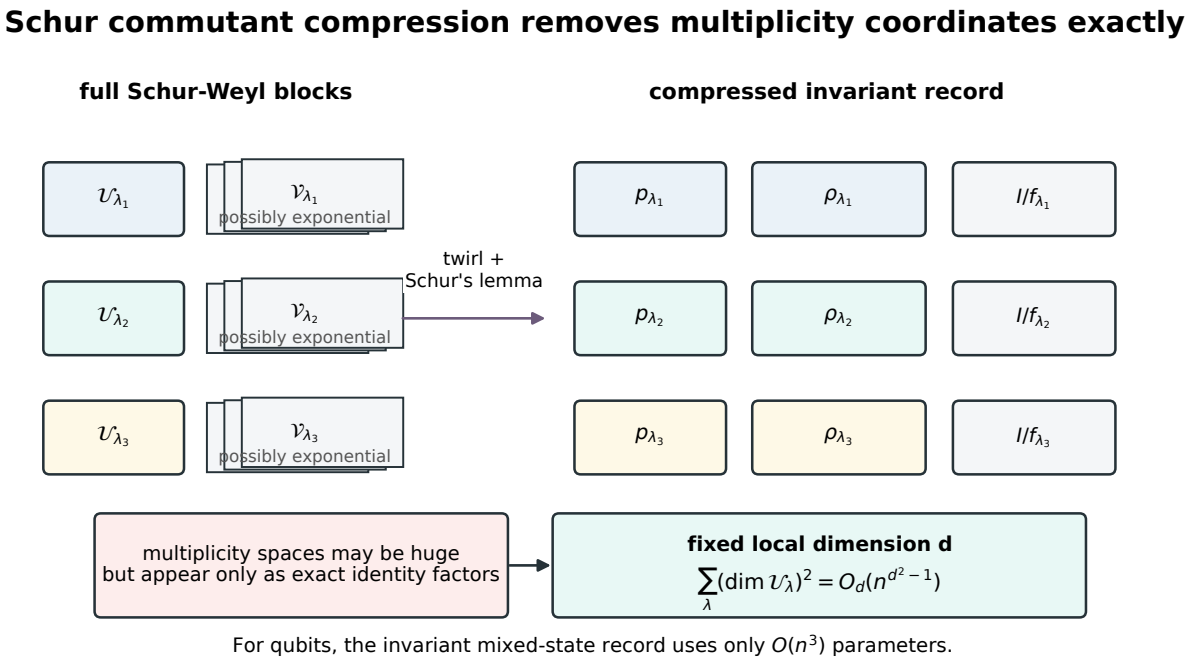


Figure 12.1: Permutation-invariant density operators retain the polynomial carrier blocks and replace each possibly exponential multiplicity space by an exact identity factor.

12.6 What this theorem does not cover

The theorem does not make an arbitrary local circuit easy. A gate acting on one selected pair generally fails to commute with all permutations. An arbitrary input may have Schur blocks that are difficult to compute from its succinct description. A non-invariant measurement can read information stored in multiplicity coordinates. Finally, allowing local dimension d to grow with n can make the commutant record superpolynomial.

The result is therefore a complete collapse theorem for a declared symmetry and access model, not a claim that Schur–Weyl decomposition alone solves generic many-body simulation.

Chapter 13

Orbit-quotient quantum walks

13.1 Orbit states

Let a group G act by automorphisms on a finite graph with vertex set V. The orbits are $\mathcal{O}_1, \ldots, \mathcal{O}_r$. Define normalized orbit states

$$|\mathcal{O}_a\rangle = \frac{1}{\sqrt{|\mathcal{O}_a|}} \sum_{x \in \mathcal{O}_a} |x\rangle.$$

If the adjacency or Hamiltonian H commutes with the group action, the span of these states is invariant whenever the partition is equitable.

The quotient matrix is

$$\widetilde{H}_{ab} = \langle \mathcal{O}_a | H | \mathcal{O}_b \rangle.$$

Theorem 13.1 (Orbit-quotient collapse)**.** *Suppose a uniformly described quantum walk has a polynomial number $r(n)$ of relevant orbits, an orbit-uniform initial state, an orbit-invariant final measurement, and polynomial-time algorithms for orbit sizes and quotient entries. Then the walk can be simulated classically in polynomial time by the $r(n) \times r(n)$ quotient dynamics.*

Proof. The orbit-state span is invariant. The state is represented by $r(n)$ amplitudes. Polynomial-time computation of quotient entries and ordinary polynomial-precision linear algebra satisfy Theorem 4.2. □

13.2 Hypercube example

The n-dimensional hypercube is invariant under coordinate permutations. Its vertex orbits are Hamming-weight layers $0, 1, \ldots, n$. In the normalized orbit basis, the quotient adjacency has couplings

$$\langle k+1 | A | k \rangle = \sqrt{(k+1)(n-k)}. \tag{13.1}$$

Thus a walk on 2^n vertices initialized at 0^n evolves in an $(n+1)$-dimensional subspace.

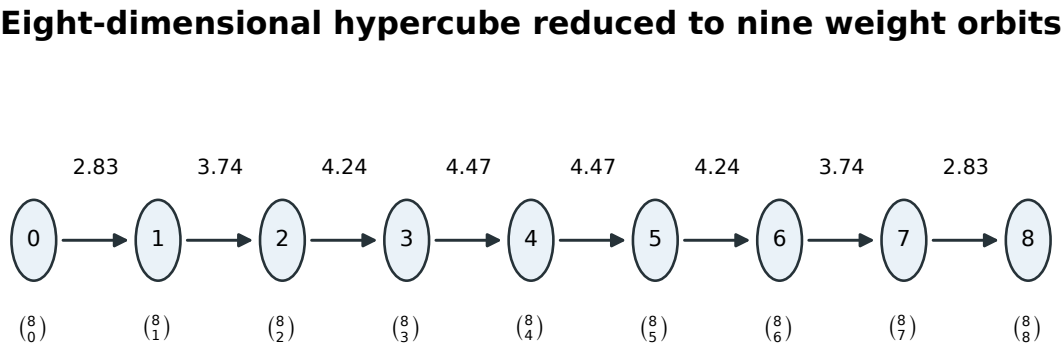


Figure 13.1: The hypercube walk collapses to a weighted path of Hamming-weight orbits.

13.3 The succinct-input caveat

If the full graph is supplied explicitly, a polynomial-in-$|V|$ algorithm may still be exponential in the succinct parameter n. The theorem requires quotient data computable directly from the succinct graph description. Discovering graph automorphisms or orbit partitions can itself be difficult. For regular families such as hypercubes, Johnson graphs, association schemes, and Cayley graphs with known character theory, the quotient is explicit.

13.4 Fourier packets

On a cycle, translation characters diagonalize the adjacency operator. The continuous-time walk

$$|\psi(t)\rangle = e^{-itA}|0\rangle$$

is a sum of Fourier-character packets. This is the same discrete Fourier transform that appears as a signal-processing coordinate change, a group-character decomposition, and a quantum basis transformation.

Chapter 14

Few bosons and few fermions

14.1 Bosonic occupation space

For N indistinguishable bosons in m modes, a Fock basis state is an occupation vector

$$(n_1, \dots, n_m), \qquad n_1 + \cdots + n_m = N.$$

The dimension is

$$D_{\text{boson}}(m, N) = \binom{N + m - 1}{N}. \tag{14.1}$$

For fixed N, this is $O(m^N)$.

14.2 Fermionic occupation space

For N fermions in m modes, each mode is occupied at most once and

$$D_{\text{fermion}}(m, N) = \binom{m}{N}. \tag{14.2}$$

For fixed N, this is also polynomial in m.

Corollary 14.1 (Few-particle collapse)**.** *A polynomial-size number-preserving bosonic or fermionic circuit with a fixed number of particles is classically polynomial-time simulable if its mode-gate updates and final observable are computable in polynomial time in the occupation basis.*

Proof. Use Equations (14.1) and (14.2) and Theorem 4.2. □

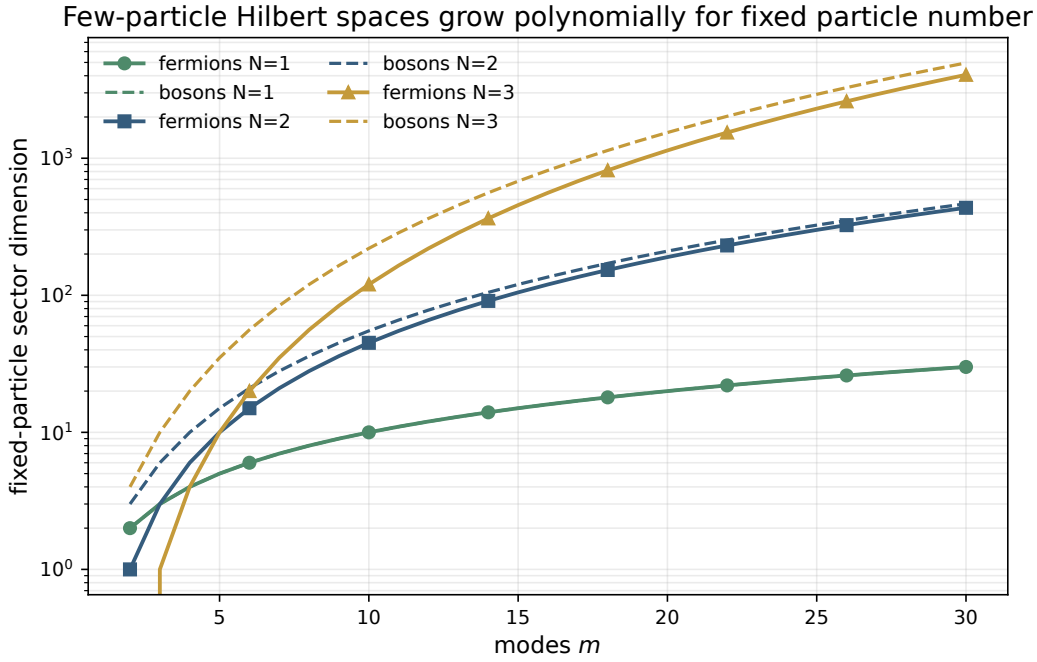


Figure 14.1: Fixed-particle bosonic and fermionic sectors grow polynomially with the number of modes.

14.3 Gate updates

A number-preserving one-body unitary acts on mode creation operators by

$$a_i^\dagger \longmapsto \sum_j U_{ji} a_j^\dagger.$$

A generic dense mode transformation can induce many occupation-basis transitions, but for fixed N the basis dimension remains polynomial. Local hopping and phase gates are sparse. Few-body interactions remain polynomial to apply for fixed N, although the exponent can grow with N.

14.4 Where the theorem stops

If N scales linearly with m, both dimensions become exponential. The fixed-number symmetry still reduces the full Fock space, but it does not prove polynomial simulation. Boson-sampling-type interference also shows that a fixed total particle number is not enough when the particle count itself scales with input size and the output asks for detailed many-body amplitudes.

Scope boundary

The corollary is a fixed-particle theorem. It is not a claim that general many-boson or many-fermion quantum dynamics are classically easy.

Chapter 15

Scaling phase diagrams for charge, particles, and local dimension

15.1 Exact sector size is a complexity phase variable

The statement "fixed particle number is polynomial" has a sharp extension. Let the particle number depend on the system size:

$$k = k(n).$$

The explicit-amplitude simulation cost of one fermionic or qubit weight packet is governed by

$$D_{n,k} = \binom{n}{k}.$$

For $1 \le k \le n$, the standard bounds

$$\left(\frac{n}{k}\right)^k \le \binom{n}{k} \le \left(\frac{en}{k}\right)^k \tag{15.1}$$

show that the growth changes continuously between polynomial and exponential regimes.

Because

$$\binom{n}{k} = \binom{n}{n-k},$$

write

$$r(n) = \min\{k(n), n - k(n)\}.$$

Theorem 15.1 (Explicit fixed-charge phase diagram)**.** *For explicit amplitude storage in a weight-$k(n)$ packet:*

1. *the packet dimension is polynomial in n if $r(n) = O(1)$;*
2. *the packet dimension is superpolynomial whenever $r(n) \to \infty$;*
3. *if $r(n) = O(\log n)$, then*
$$D_{n,k} = 2^{O((\log n)^2)},$$
so the explicit packet simulation is quasipolynomial;
4. *if $r(n) = O(n^\alpha)$ for a fixed $0 < \alpha < 1$, then*
$$D_{n,k} = 2^{O(n^\alpha \log n)},$$
which is subexponential;

5. *if* $k(n)/n \to p \in (0,1)$*, then*

$$\log_2 D_{n,k} = nH_2(p) + O(\log n),$$

where H_2 *is binary entropy, and the packet remains exponential.*

Proof. The upper bounds follow from Equation (15.1). If $r = O(1)$, then $\binom{n}{r} = O(n^r)$. Conversely, suppose $r \to \infty$ and $r \le n/2$. If $r \le \sqrt{n}$, then

$$\log \binom{n}{r} \ge r \log(n/r) \ge \frac{r}{2} \log n,$$

whose ratio to $\log n$ diverges. If $r > \sqrt{n}$, then

$$\log \binom{n}{r} \ge r \log 2 \ge \sqrt{n} \log 2,$$

which also dominates every constant multiple of $\log n$. The linear-density estimate is the usual entropy asymptotic for binomial coefficients. □

Derived theorem or corollary

For the explicit occupation-amplitude representation, "few particles" means a bounded number of particles or holes. A slowly growing charge can still improve on full state-vector scaling, but it no longer guarantees membership in P.

15.2 A bounded-excitation hierarchy

The union of the first K weight sectors has dimension

$$D_{\le K}(n) = \sum_{k=0}^{K} \binom{n}{k}.$$

For fixed K,

$$D_{\le K}(n) = O(n^K).$$

If $K = O(\log n)$, the same union is quasipolynomial. This refines Theorem 6.6: bounded symmetry breaking preserves polynomial simulation, while logarithmically growing excitation gives a controlled quasipolynomial regime.

15.3 Bosonic and fermionic analogues

For N fermions in m modes,

$$D_{\mathrm{F}}(m,N) = \binom{m}{N}.$$

For N bosons in m modes,

$$D_{\mathrm{B}}(m,N) = \binom{m+N-1}{N}.$$

The same binomial analysis gives:

Corollary 15.2 (Few-particle scaling)**.** *With the number of modes as the input-size parameter:*

1. *fixed N gives polynomial explicit-amplitude simulation;*
2. *$N = O(\log m)$ gives quasipolynomial sector size;*
3. *$N = O(m^\alpha)$, $0 < \alpha < 1$, gives subexponential sector size;*
4. *finite-density fermionic sectors remain exponential.*

For bosons, the statements hold in the dilute regime with the same occupation-basis access assumptions.

The adjective "polynomial" here refers to the sector dimension and efficient gate action. It does not assert that every interacting Hamiltonian has efficiently computable matrix elements or that every compact description remains closed.

15.4 Symmetric qudits with growing local dimension

The symmetric d-level subspace has dimension

$$S(n, d) = \binom{n + d - 1}{d - 1}.$$

For fixed d, this is polynomial in n. If $d = d(n)$ grows, the same phase transition appears.

Proposition 15.3 (Growing-local-dimension boundary)**.** *In the explicit occupation basis:*

1. *fixed d gives $S(n, d) = O(n^{d-1})$;*
2. *$d = O(\log n)$ gives $S(n, d) = 2^{O((\log n)^2)}$;*
3. *unbounded d is not covered by the fixed-d polynomial-collapse theorem without an additional internal compression.*

Thus permutation symmetry removes the exponential tensor-product multiplicity only when the number of local levels is controlled, or when the occupation record itself admits a second compression.

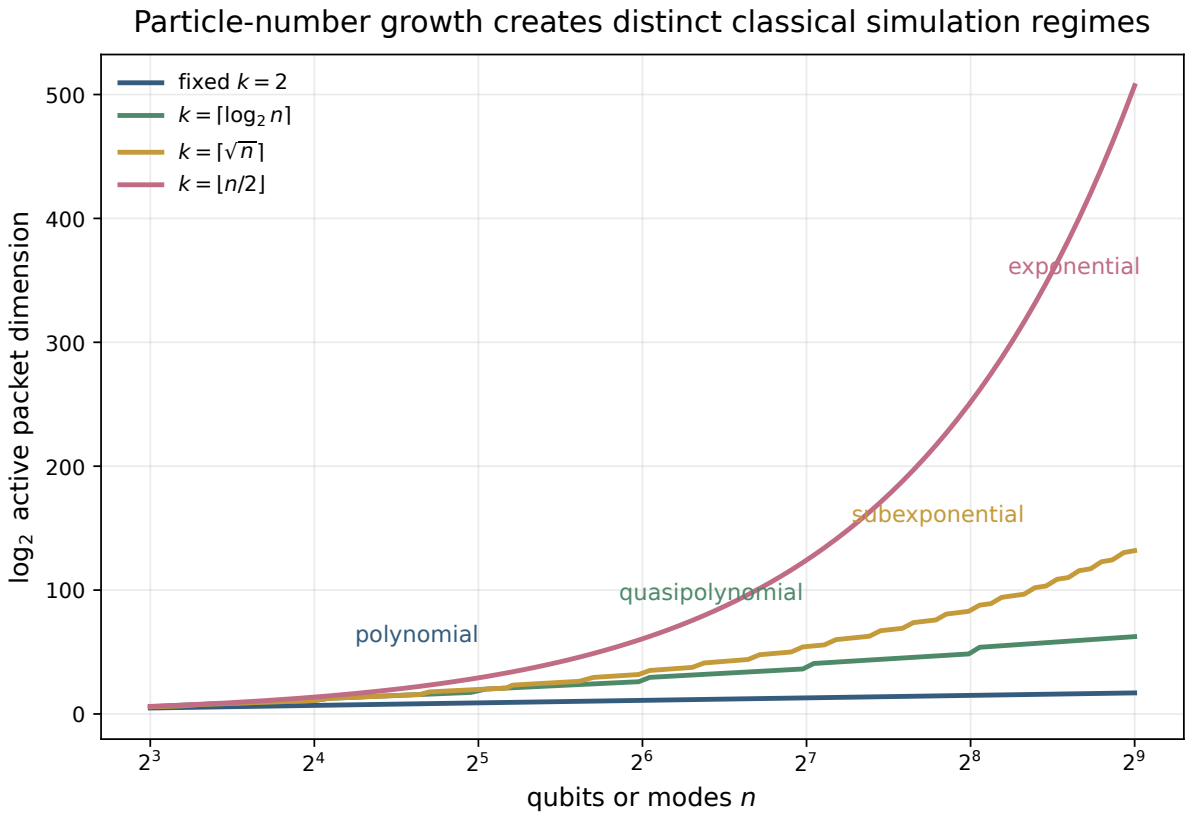


Figure 15.1: The active dimension passes through polynomial, quasipolynomial, subexponential, and exponential regimes as particle number grows.

15.5 Complexity class versus practical runtime

A polynomial sector can still have a large exponent. Fixed $k = 20$ gives $O(n^{20})$ explicit amplitudes, which is in P but may be impractical. Conversely, a subexponential sector may be useful at moderate size without changing the formal complexity class.

A complete performance claim should therefore report both:

asymptotic class and actual representation size and update cost.

This is why the benchmark chapters keep wall-clock evidence separate from the dimension theorems.

Chapter 16

Stabilizer syndromes and Clifford packets

16.1 Syndrome projectors

Let $S_1, \dots, S_m$ be commuting Pauli stabilizers with eigenvalues ± 1. A syndrome $s \in \{0,1\}^m$ defines

$$P_s = \prod_{j=1}^{m} \frac{I + (-1)^{s_j} S_j}{2}.$$

These projectors are orthogonal and sum to the identity. Syndrome measurement selects one packet, while recovery applies a syndrome-dependent transfer toward the code packet.

16.2 The syndrome label is not the logical state

If the code encodes k logical qubits, a fixed syndrome packet can have dimension 2^k. The syndrome has only m bits, but arbitrary logical information remains inside the packet. Syndrome packetization therefore does not by itself prove efficient simulation.

16.3 Tableau compression

Clifford circuits provide a different route. A stabilizer state may occupy exponentially many computational-basis strings, yet it is represented by a polynomial-size stabilizer tableau. Clifford gates update the tableau in polynomial time, and Pauli measurements are efficiently computable. This is an example of the compressed-packet theorem rather than a small-dimensional packet theorem.

Derived theorem or corollary

Packet labels and internal encoding are complementary. A Clifford circuit restricted to selected syndromes may be simulated by a tableau augmented with the active syndrome graph, even when the packet dimensions are exponential.

16.4 Non-Clifford residues

A few non-Clifford gates can be handled by stabilizer decompositions, quasiprobability expansions, or magic-state rank methods. The cost grows with a non-Clifford resource such as stabilizer rank or

robustness. Symmetry packets may reduce the number of relevant branches, but they do not eliminate the intrinsic cost of the non-Clifford residue.

16.5 Error-correction interpretation

The packet view separates three objects:

$$\text{syndrome label}, \qquad \text{logical state}, \qquad \text{recovery transfer}.$$

This separation is useful for compressed simulation of fault-tolerant protocols, but a full complexity statement must include repeated measurements, classical control, and the size of the logical state.

Chapter 17

Gaussian, fermionic, and matchgate compression

17.1 Polynomial covariance representations

Gaussian bosonic and fermionic states are specified by first and second moments rather than by all Fock-basis amplitudes. A bosonic covariance matrix has size polynomial in the number of modes, and a fermionic Gaussian state is represented by an antisymmetric Majorana covariance matrix. Gaussian unitaries act by symplectic or orthogonal transformations:

$$\Gamma \longmapsto S\Gamma S^T \quad \text{or} \quad \Gamma \longmapsto O\Gamma O^T.$$

This is a compact internal encoding even when the corresponding Hilbert-space packet is exponentially large.

17.2 Hybrid symmetry and Gaussianity

Number conservation, parity, translation symmetry, and point-group symmetry can further block the covariance matrices. A simulator can therefore combine two reductions:

$$\text{symmetry packetization} + \text{Gaussian internal encoding}.$$

The first removes unreachable sectors. The second compresses states within the remaining sectors.

17.3 Matchgate circuits

Nearest-neighbor matchgate circuits have polynomial classical representations through fermionic Gaussian dynamics. Geometry matters: adding SWAP, fermionic SWAP, long-range gates, or a non-matchgate resource changes the simulation problem. A symmetry-collapse atlas should therefore record the gate library and connectivity as part of the model, not as implementation details.

17.4 Low-rank non-Gaussian corrections

One research direction is to represent a weakly non-Gaussian state as

$$\rho = \sum_{a=1}^{R} c_a \rho_a^{\text{Gauss}}$$

with small Gaussian rank R, or as a Gaussian reference plus sparse packet corrections. A polynomial collapse requires R and the update cost to remain polynomial.

Open program or boundary

Gaussianity is an internal representation class, not a symmetry projector. A complete hybrid theorem must prove closure of the chosen compressed class under every allowed gate and bound the growth of any non-Gaussian residue.

17.5 Few-particle versus Gaussian routes

Fixed-particle simulation and Gaussian simulation are independent. A fixed two-fermion state may be non-Gaussian but live in a polynomial-dimensional sector. A half-filled Gaussian state may live in an exponential number sector but have a polynomial covariance description. The two-axis model in Chapter 22 is designed to capture this distinction.

Part III

States, measurements, and open dynamics

Chapter 18

Density operators: populations and coherences

18.1 Two packet indices

A pure state uses one packet label. A density operator uses two:

$$\rho = \sum_{\alpha,\beta} P_\alpha \rho P_\beta.$$

The diagonal block $P_\alpha \rho P_\alpha$ is population inside packet α. The off-diagonal block $P_\alpha \rho P_\beta$ is coherence between packets α and β.

Discarding off-diagonal blocks is not merely a change of coordinates. It is a physical operation such as dephasing. An exact simulator must preserve all coherence blocks relevant to the final observable.

18.2 Hilbert–Schmidt decomposition

Orthogonality of the left-right operator packets gives

$$\|\rho\|_2^2 = \sum_{\alpha,\beta} \|P_\alpha \rho P_\beta\|_2^2.$$

This identity supplies an exact coherence-energy ledger. It can be used to decide which blocks are large, to audit dephasing, or to bound truncation errors in approximate packet simulations.

18.3 Symmetry-preserving unitary dynamics

If U commutes with every packet projector, then

$$P_\alpha U \rho U^\dagger P_\beta = U(P_\alpha \rho P_\beta)U^\dagger.$$

The left and right packet labels are preserved. Populations and coherences evolve independently by their declared labels.

18.4 Observable sparsity

For an observable M,

$$\mathrm{Tr}(M\rho) = \sum_{\alpha,\beta} \mathrm{Tr}(P_\beta M P_\alpha \, P_\alpha \rho P_\beta).$$

Only operator packets coupled by M contribute. A measurement that is diagonal in the symmetry labels depends only on populations. A phase-sensitive measurement can require off-diagonal coherences.

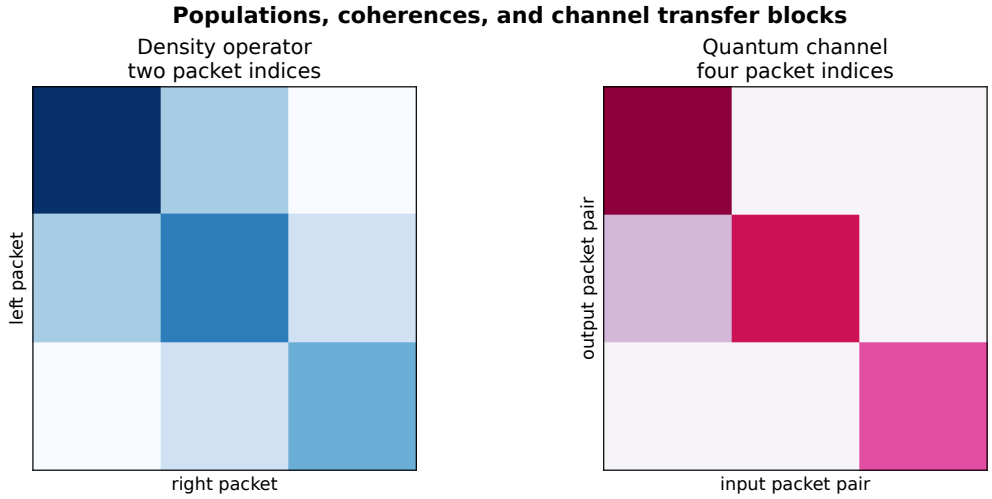


Figure 18.1: State operators carry two packet indices, while channels act between pairs of packet labels.

18.5 Polynomial operator-space collapse

If only polynomially many density blocks are reachable and every block has a polynomial representation, then mixed-state evolution can also be simulated in polynomial time. The representation size is not the square of the full Hilbert dimension. It is the total cost of the active left-right blocks.

Derived theorem or corollary

For a polynomial active set A_n of packet labels, a density-matrix simulator has at most $|A_n|^2$ label pairs. Polynomial-time simulation follows when each active coherence block has a polynomial classical representation and channel updates remain sparse and efficient.

Chapter 19

Channels, Kraus maps, and Lindblad generators

19.1 Four-index transfer

A quantum channel with Kraus operators K_μ acts by

$$\Phi(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger.$$

Relative to operator packets

$$\mathcal{P}_{\alpha\beta}(X) = P_\alpha X P_\beta,$$

the channel has blocks

$$\Phi_{\gamma\delta,\alpha\beta} = \mathcal{P}_{\gamma\delta}\,\Phi\,\mathcal{P}_{\alpha\beta}.$$

Thus a channel carries four packet labels: an input coherence pair and an output coherence pair.

19.2 Symmetry-preserving channels

If every Kraus operator commutes with the symmetry action, the channel is diagonal in the isotypic labels. More generally, Kraus operators may carry definite charges, producing a sparse selection rule. Amplitude damping transfers population between number sectors. Dephasing damps selected off-diagonal blocks. Particle loss moves weight-k population to weight $k-1$.

19.3 Lindblad dynamics

A Markovian open system obeys

$$\frac{d\rho}{dt} = \mathcal{L}(\rho) = -i[H,\rho] + \sum_a \left(L_a \rho L_a^\dagger - \frac{1}{2}\{L_a^\dagger L_a, \rho\} \right).$$

Packet selection rules of H and L_a determine a sparse directed graph on coherence pairs. If the reachable graph and each block representation remain polynomial, a classical integrator can evolve the reduced system.

19.4 Error accounting for truncated channels

Let Π_A project operator space onto an active set of coherence blocks. A reduced evolution uses

$$\mathcal{L}_A = \Pi_A \mathcal{L} \Pi_A.$$

The discarded transfer is

$$(I - \Pi_A)\mathcal{L}\Pi_A.$$

Duhamel's formula yields an error bound analogous to the Hamiltonian leakage theorem. The resulting estimate depends on the induced norm of the discarded superoperator transfer and the integration time.

19.5 Syndrome-conditioned recovery

Error correction naturally fits the four-index model. Syndrome measurement selects a population packet, classical processing chooses a recovery, and the recovery transfers the state toward the code packet. A full simulation must still retain the logical state inside each syndrome sector.

Open program or boundary

A useful open-system collapse theorem should state the norm, time horizon, packet-discovery algorithm, integration precision, and growth of reachable coherence pairs. Sparsity of a displayed block matrix is not enough by itself.

Chapter 20

Measurement, conditioning, and classical feedback

20.1 Projective measurement

For projectors $\{P_\alpha\}$, the probability of outcome α is

$$p_\alpha = \mathrm{Tr}(P_\alpha \rho),$$

and the conditioned state is

$$\rho_\alpha = \frac{P_\alpha \rho P_\alpha}{p_\alpha}$$

when $p_\alpha > 0$. Measurement is therefore packet selection followed by normalization.

20.2 Branching complexity

A measurement-assisted circuit produces a tree of outcomes. Classical simulation is efficient when the branch distribution can be sampled or summed without enumerating exponentially many relevant histories. Symmetry can merge histories that lead to the same packet state, but only if the feedforward policy respects the same equivalence.

20.3 Dynamic programming on packet states

Suppose each branch is represented by a pair (α, r), where α is a packet label and r is a compact internal state. If polynomially many distinct pairs occur at every depth, identical pairs can be merged and their probabilities added. This gives a dynamic program instead of an exponential branch tree.

Proposition 20.1 (Branch-merging simulation)**.** *If a measurement-and-feedback circuit has polynomial depth, polynomially many distinct reachable packet representations at each step, polynomial-time transition probabilities, and an efficient final predicate, then its output distribution is classically computable in polynomial time.*

Proof. Maintain a probability dictionary over distinct packet representations. Each measurement or controlled gate updates every entry into polynomially many successors. Merge equal successors. The dictionary remains polynomial by hypothesis. □

20.4 Reset and dissipation

Reset is nonunitary but often simplifies a classical representation by removing entropy or closing branches. Conversely, measurement may expose a classical label while leaving a complex logical state. The resource value of measurement depends on the target, ancilla convention, and cost function.

20.5 Connection to catalytic gate models

The larger gate-set monograph studies temporary excursions into measurement, ancilla, or symmetry-breaking resources. The present monograph asks a different question: when does the resulting model remain classically simulable? A temporary resource may shorten a quantum circuit without making the computation classically easy.

Chapter 21

Block encodings and polynomial operator transforms

21.1 Projector block form

A block encoding places a generally nonunitary operator A inside a larger unitary U so that

$$(\langle 0| \otimes I)U(|0\rangle \otimes I) = \frac{A}{\alpha}.$$

With

$$P_0 = |0\rangle\langle 0| \otimes I,$$

this is the selected block

$$P_0 U P_0 = \frac{A}{\alpha}$$

after identifying the encoded subspace.

21.2 Structural connection

The encoded operator is literally a projector block of a unitary. Polynomial transformations of singular values can be implemented by controlled phase sequences once the block encoding and access model are available. This makes block encodings a natural part of a projector calculus.

21.3 Why block encoding does not imply classical simulation

A block encoding can be efficient on a quantum computer because U is supplied by an oracle or a succinct circuit. A classical algorithm may not have efficient access to the entries of A, to the block state, or to the polynomial transform. The selected block may still have exponential dimension.

Scope boundary

The statement "A is a block of U" is structural. A classical collapse requires a polynomial classical description of the active encoded state, efficient application of the selected block, and efficient evaluation of the output.

21.4 Symmetry-resolved block encodings

If A commutes with a symmetry group, both the encoded system and the ancilla can be decomposed into packets. A symmetry-resolved block encoding may reduce normalization, memory, or simulation cost for selected sectors. The relevant objects are

$$P_\chi P_0 U P_0 P_\psi.$$

Exact symmetry removes off-diagonal $\chi \neq \psi$ blocks. Approximate symmetry yields leakage terms that must be included in the error budget.

21.5 Research direction

A useful classical theorem would identify access models in which the active block has polynomial packet dimension or a compact internal representation. This could cover structured sparse matrices, low-rank updates, association-scheme operators, and fixed-particle Hamiltonians.

Part IV

Complexity boundary

Chapter 22

Compressed packet simulation

22.1 Beyond packet dimension

The small-packet theorem stores one amplitude per packet-basis state. This is not the only classical representation. A state in an exponentially large packet may lie on a polynomially parameterized manifold. Stabilizer states, Gaussian states, matrix-product states of bounded bond dimension, and low-rank operator states are standard examples of compact internal encodings.

Let $\mathcal{R}_n$ be a class of classical descriptions, and let

$$\mathrm{decode} : \mathcal{R}_n \to P_n\mathcal{H}_n$$

associate a quantum state with each description.

Theorem 22.1 (Compressed-packet simulation)**.** *A polynomial-time uniform quantum circuit family lies in P if there is a polynomial-size classical representation satisfying:*

1. *the initial state has an efficiently constructible description $r_0 \in \mathcal{R}_n$;*
2. *every gate induces a polynomial-time update $r \mapsto T_j(r)$ with*

$$\mathrm{decode}(T_j(r)) = G_j\,\mathrm{decode}(r);$$

3. *all intermediate description sizes remain polynomial;*
4. *the acceptance probability is computable from the final description in polynomial time.*

Proof. Iterate the polynomial-time classical updates on the polynomial-size descriptions and evaluate the final measurement. The algorithm never reconstructs the full state vector. □

22.2 Two-axis complexity

The records suggest the pair

$$(D_{\mathrm{packet}}, \chi_{\mathrm{internal}}),$$

where D_{packet} measures active packet size or count and χ_{internal} measures the representation cost inside the packet. Efficient simulation may follow from either coordinate:

- fixed weight has polynomial D_{packet} and can use explicit amplitudes;
- stabilizer states may have exponential support but polynomial χ_{internal};

- Gaussian states may occupy large number sectors but have polynomial covariance descriptions;
- half-filled generic states may have both coordinates exponential.

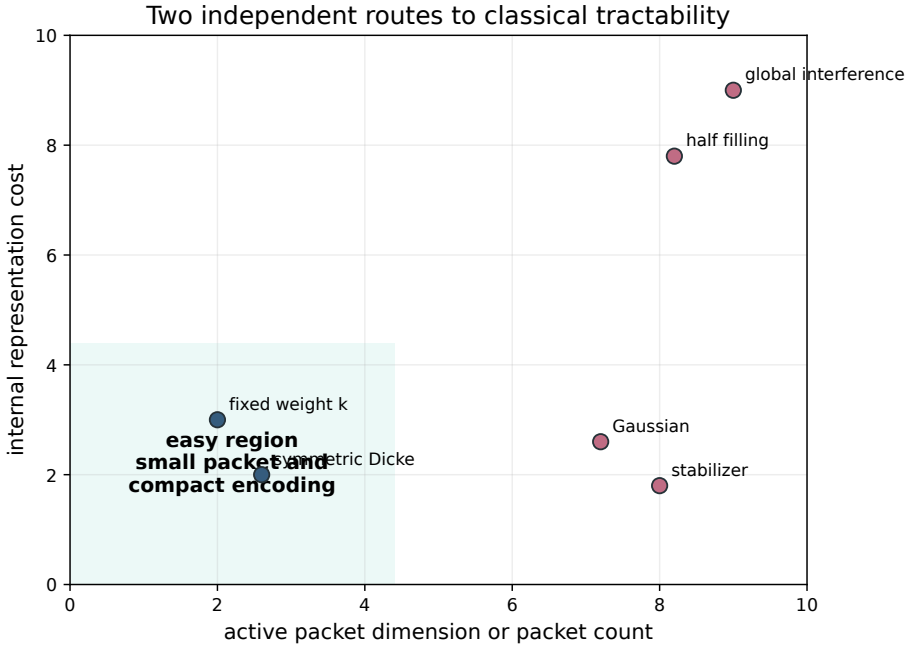


Figure 22.1: Packet dimension and internal representation cost are independent classical resources.

22.3 Hybrid transfer graphs

Suppose packets are nodes of a sparse graph and each node carries an internal representation. A gate may update a node internally or transfer a compact component to neighboring nodes. The simulation cost depends on

$$\sum_{\alpha \in A_t} \mathrm{size}(r_\alpha) + \#\{\text{nonzero transfers at time } t\}.$$

This model combines symmetry selection rules with compact state classes.

22.4 Closure is the decisive condition

A compact representation may be destroyed by one gate. Clifford tableaux fail under generic non-Clifford rotations. Gaussian covariance matrices fail under generic non-Gaussian operations. Low-bond tensor networks may acquire exponential bond dimension. The theorem therefore requires closure or a proved polynomial growth bound, not merely a compact initial state.

Principle 22.2 (Representation growth audit)**.** *The correct complexity variable is the maximum representation size reached along the computation, not the size of the initial or final state alone.*

Chapter 23

Global interference and the failure of local packet collapse

23.1 A different quantum mechanism

Invariant-packet reduction succeeds when large parts of Hilbert space are never occupied or can be compressed locally. A different regime stores the answer in coherent relations among many paths. The individual amplitudes may be simple, while their signed sum is the computational object.

23.2 Forrelation amplitude

Let $f, g : \{0,1\}^n \to \{-1,1\}$ and $N = 2^n$. The circuit with alternating Hadamards and phase oracles has zero-state amplitude

$$\Phi(f,g) = \frac{1}{N^{3/2}} \sum_{x,y} f(x)(-1)^{x \cdot y} g(y). \tag{23.1}$$

The circuit does not evaluate the N^2 terms separately. It preserves phases and lets them interfere.

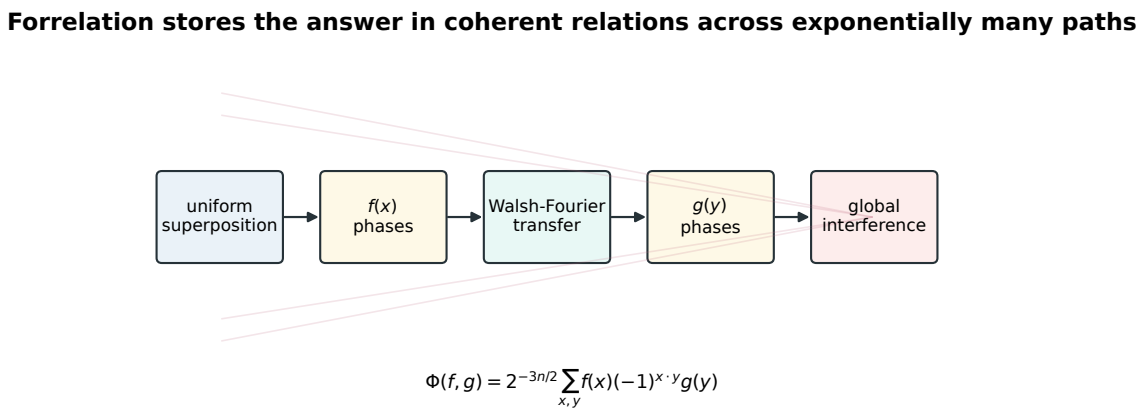


Figure 23.1: Forrelation is governed by coherent Fourier transfer rather than a small invariant sector.

23.3 Why Fourier structure is not automatically easy

The Walsh transform is explicit and highly symmetric. That does not imply that the bilinear expression in Equation (23.1) is classically polynomial-time computable when f and g are supplied through restricted query access. The difficulty lies in the access model and the coherent global relation, not in the absence of algebraic structure.

23.4 Truth tables versus succinct access

If the complete truth tables of f and g are part of the classical input, their length is already N, and an $O(N \log N)$ Walsh transform is polynomial in the explicit input length. In a query or succinct-circuit model, n is the parameter and $N = 2^n$. The same formula then represents exponentially many values.

This distinction illustrates a general rule: complexity statements must be measured against the input encoding. A quotient of exponential dimension may be efficient relative to an explicit matrix input but exponential relative to a succinct circuit input.

23.5 Cross-packet coherence

Suppose a projector family gives many packets P_α. Even if each diagonal block is simple, the output can depend on the phases of

$$P_\alpha|\psi\rangle \quad \text{relative to} \quad P_\beta|\psi\rangle$$

for exponentially many pairs. A simulator that retains only populations

$$\|P_\alpha|\psi\rangle\|^2$$

loses this information. The hard-side invariant must measure classical compression of coherent relations, not merely active dimension.

Open program or boundary

A representation-independent lower bound on interference compression for an explicit, succinct, uniform circuit family would be a major complexity result. The packet framework states the missing object but does not provide that lower bound.

Chapter 24

Simulation complexity versus decision complexity

24.1 Why simulator lower bounds are insufficient

Showing that a circuit defeats one simulation strategy does not prove that its decision problem is outside P. A classical algorithm may compute the answer without reconstructing the quantum state. It is therefore insufficient to rule out only:

- full state-vector simulation,
- one projector family,
- bounded-width tensor-network contraction,
- path sampling,
- low-rank transfer matrices,
- stabilizer decompositions,
- any fixed catalogue of known simulators.

A proof of $P \neq BQP$ would need a lower bound against every deterministic polynomial-time classical algorithm for an explicit language.

24.2 A representation-relative invariant

For a declared classical representation class $\mathcal{R}$, define

$$\mathrm{TC}^{\mathcal{R}}_{\varepsilon}(C) = \min\left\{\mathrm{cost}(R) : R \in \mathcal{R},\ |R(C) - p_C| \leq \varepsilon\right\},$$

where p_C is the target acceptance probability. This permits a disciplined ladder:

1. exact polynomial-packet simulation for easy sectors;
2. lower bounds for local packet-query methods;
3. lower bounds for broader tensor, sampling, low-degree, and communication representations;
4. a representation-independent lower bound;
5. a separation of P and BQP.

Only the first step is completed here.

24.3 Universal packetization and its consequence

Consider the statement:

Conjecture 24.1 (Universal polynomial packetization). Every polynomial-time uniform quantum circuit family admits an efficiently discoverable exact or approximate packet representation of polynomial active size and polynomial transfer complexity, sufficient to compute its acceptance probability.

If true, Theorems 4.2 and 22.1 would imply $BQP \subseteq P$, and hence $P = BQP$. The records do not establish this conjecture and explicitly identify global interference as an obstacle.

24.4 The opposite route

An explicit uniform family with a proved representation-independent classical lower bound and a polynomial quantum algorithm would establish $P \neq BQP$. The candidate succinct transfer-interference problem in the records illustrates the form of such a target, but the required lower bound is the unresolved theorem itself.

Scope boundary

This monograph proves polynomial classical simulation for declared subclasses. It neither proves nor assumes a universal collapse of quantum computation.

24.5 Nonrelativizing issues

Oracle separations show that query complexity can distinguish quantum and classical access. They also show why a black-box argument alone cannot settle ordinary P versus BQP. A full separation would have to use explicit features such as uniform circuit generation, arithmetic structure, algebraic degree, communication constraints, or another nonrelativizing mechanism.

Chapter 25

Existence defects and quantum combinatorial realizations

25.1 Metric collapse and existence collapse

The simulation theorems compare costs of two realizations that both exist. A broader restricted-realization framework also allows the restricted object to be absent. If

$$\mathcal{R}(x) = \varnothing, \qquad \mathcal{A}(x) \neq \varnothing,$$

then enlarging the realization category changes feasibility rather than merely runtime.

25.2 Quantum Latin-square example

The order-six quantum-Latin-square record supplies an existence-level example. Entangled quantum Latin squares of order six exist, while mutually orthogonal non-entangled quantum Latin squares of order six do not. This is an ambient-versus-restricted realization gap, not a proof of classical simulation and not a gate-count shortcut.

The proof architecture uses support patterns, unitary-pattern constraints, graph orthonormal representations, projective-line closure, dimension contradictions, and a finite reduction to twelve classical Latin-square classes. Its conceptual relevance here is limited but useful: enlarging the allowed quantum resource can change whether a structure exists at all.

25.3 Why this does not imply computational advantage

Existence of an entangled combinatorial design does not by itself provide an algorithmic speedup. One must still specify an input problem, circuit family, output rule, and classical comparison. Likewise, nonexistence of a product realization does not prove classical hardness of the entangled realization.

Scope boundary

The quantum Latin-square result is included as a contextual example of restricted-realization failure. It is not used as a premise in any packet-simulation theorem.

25.4 Unified terminology

For a target x, let $c_{\mathrm{res}}(x)$ and $c_{\mathrm{amb}}(x)$ be minimum costs, with $+\infty$ assigned when a realization does not exist. Then

$$\Delta(x) = c_{\mathrm{res}}(x) - c_{\mathrm{amb}}(x)$$

includes both finite metric defects and infinite existence defects. Symmetry collapse concerns the special case where the ambient quantum description has a cheaper classical realization because its reachable representation is compressed.

Part V

Algorithms and research program

Chapter 26

Architecture of a symmetry-collapse compiler

26.1 The five-stage pipeline

A practical compiler must perform more than symbolic block diagonalization.

1. Detect exact or approximate symmetries.
2. Construct an invariant basis, orbit quotient, or compact internal representation.
3. Compile each gate into that representation.
4. Track leakage, truncation, and numerical error.
5. Evaluate the requested measurement and emit a complexity certificate.

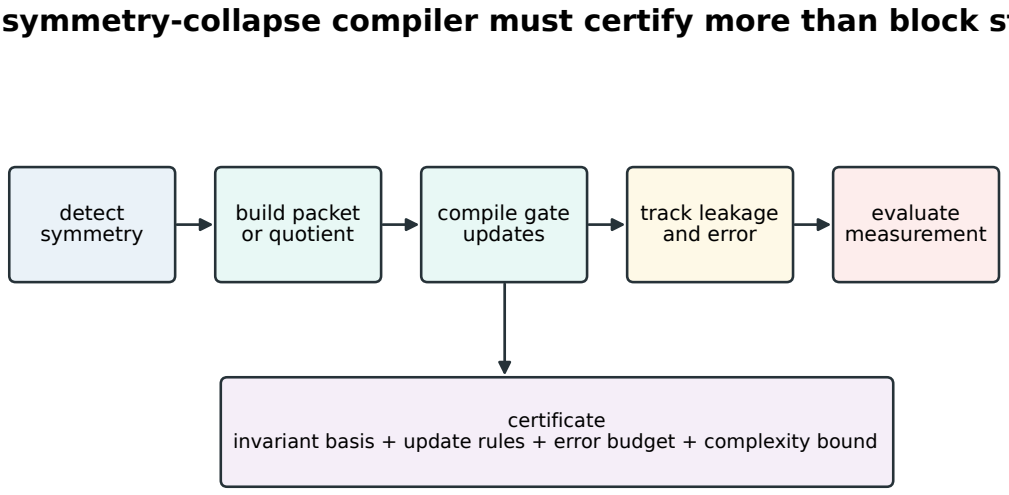


Figure 26.1: A symmetry-collapse compiler and its required certificate.

26.2 Symmetry detection

Detection may use declared conserved quantities, commutators, finite-group generators, graph automorphisms, stabilizer relations, occupation conservation, or numerical approximate symmetries. Exact symbolic detection is preferable when available. Numerical detection must specify tolerances and how they propagate.

26.3 Basis construction

Examples include:

- subset indexing for fixed weight,
- occupation vectors for bosons,
- Dicke states for full permutation symmetry,
- Schur coordinates for broader permutation symmetry,
- orbit states for graph walks,
- stabilizer tableaux,
- fermionic or bosonic covariance matrices,
- matrix-product states for bounded entanglement.

The compiler should avoid materializing an exponentially large change-of-basis matrix. It needs direct indexing and update formulas.

26.4 Gate plan caching

For a repeated circuit family, compile a packet plan containing affected index pairs, sparse matrix blocks, phase tables, and quotient couplings. The plan can be reused across parameter values when the sparsity pattern is fixed.

26.5 Fallback and adaptive enlargement

If a gate leaks from the current representation, the compiler can reject the reduction, merge packets, add neighboring sectors, or switch internal representations. The choice must preserve a certified error budget.

26.6 Output certificate

A useful machine-readable certificate should include:

- the symmetry or invariant-subspace proof,
- active dimensions or representation sizes,
- gate update complexity,
- leakage or approximation bounds,
- measurement complexity,

- benchmark and hardware metadata,
- the final complexity conclusion and its scope.

Chapter 27

Benchmark and validation protocol

27.1 Fair comparison

A packet simulator and a full simulator should use the same precision, gate parameters, initial state, observable, and arithmetic path wherever possible. Report compilation separately from repeated application.

27.2 Required metrics

A complete benchmark should report:

1. full and reduced dimensions,
2. packet-discovery and compilation time,
3. one-shot runtime,
4. repeated-application runtime,
5. peak memory,
6. reconstruction or observable error,
7. symmetry leakage,
8. hardware and software versions,
9. number of repetitions over which compilation is amortized.

27.3 Exact validation path

For manageable n, embed a random packet state into the full space, apply identical gates in both representations, and compare final states. The included validation program performs this check for exact fixed-weight circuits.

27.4 Scaling studies

A scaling plot should include both fine packets and coarse packets. Fixed weight two demonstrates polynomial dimension. Parity demonstrates a constant-factor packet. Half filling demonstrates an

exponential packet with a polynomial memory reduction factor. These controls prevent overinterpreting a speedup as a complexity collapse.

27.5 Leakage tests

Deliberately append a gate that breaks the proposed symmetry and verify that the simulator detects nonzero leakage. The archived tests break exact weight, parity, and weight modulo four, producing retained probabilities below one. A simulator that reports zero error after a symmetry-breaking gate without enlarging its representation is incorrect.

27.6 Reproducibility contract

The release preserves:

- the benchmark JSON data,
- the Python figure generator,
- packet and gate utilities,
- exact validation tests,
- LaTeX source,
- deterministic build commands,
- checksums of final artifacts.

Recorded exact computation

The release validation reproduces fixed-weight basis dimensions, residue partitions, symmetric/bosonic/fermionic dimension formulas, hypercube quotient symmetry, and full-vs-packet state reconstruction near machine precision.

Chapter 28

Atlas of candidate symmetry collapses

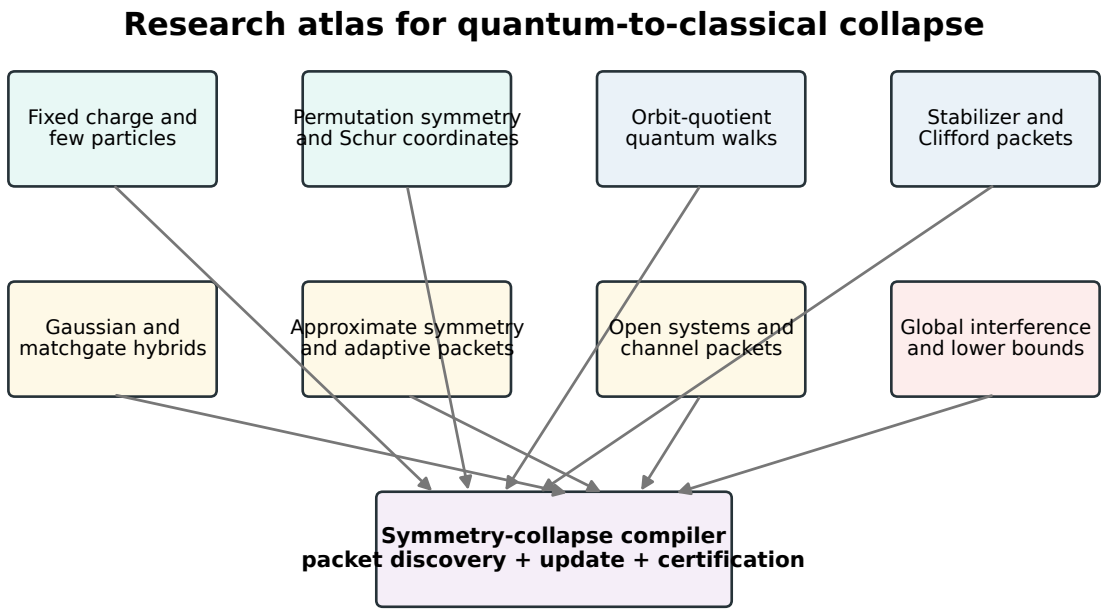


Figure 28.1: The main theorem and implementation directions organized around one certified compiler.

28.1 High-confidence theorem families

Table 28.1: High-priority collapse families.

Family	Compression mechanism	Decisive condition
Fixed charge $k = O(1)$	Explicit subset basis of size $\binom{n}{k}$	Number-preserving gates and efficient measurement
Permutation-symmetric qubits	Dicke basis of size $n + 1$	Collective or permutation-invariant dynamics

Family	Compression mechanism	Decisive condition
Symmetric qudits, fixed d	Occupation basis $\binom{n+d-1}{d-1}$	Efficient occupation-space updates
Polynomial-orbit walks	Quotient Hamiltonian on orbit states	Succinct quotient entries and orbit-invariant input/output
Few bosons or fermions	Fixed-particle occupation basis	Fixed particle number and efficient few-body updates
Stabilizer circuits	Tableau representation	Clifford closure and efficient Pauli measurement
Gaussian circuits	Covariance representation	Gaussian closure and efficient observable evaluation

28.2 Hybrid theorem targets

The strongest new classes may combine several resources:

- charge packets plus Gaussian covariance,
- permutation packets plus bounded-bond tensor networks,
- stabilizer tableaux plus a small syndrome transfer graph,
- orbit quotients plus Fourier diagonalization,
- adaptive approximate packets plus rigorous leakage accounting.

28.3 Negative controls

A research atlas should preserve cases that do not collapse:

- parity alone,
- weight modulo a fixed constant,
- half filling with generic internal states,
- a syndrome label with exponentially many logical qubits,
- a block encoding without classical access,
- global phase interference without a compact representation.

28.4 Priority order

The recommended sequence is:
1. implement the Dicke and orbit-quotient simulators;
2. combine fixed charge with Gaussian and stabilizer encodings;
3. develop adaptive leakage-certified packet growth;
4. build open-system four-index packet compilers;
5. search for broader representations of global interference;
6. separate simulator-specific lower bounds from decision-problem lower bounds.

Chapter 29

Open problems

29.1 Efficient hidden-symmetry discovery

Question 29.1. Which uniform circuit families have polynomial-dimensional invariant subspaces that can be discovered in polynomial time from the circuit description?

A nonconstructive invariant subspace does not yield an algorithm. The problem includes symbolic commutant computation, graph-orbit discovery, representation decomposition, and approximate invariant-subspace detection.

29.2 Moving-frame discovery

Question 29.2. Which circuit families admit efficiently computable frames W_t that expose one fixed polynomial packet even though no fixed laboratory-frame symmetry is preserved?

The theorem is exact, but the discovery problem is the substance. Useful frames must have compact descriptions and restricted gate matrices that can be generated without solving the original simulation problem.

29.3 Permutation-commutant compilers

Question 29.3. For which succinct permutation-invariant Hamiltonians and channels can all Schur-commutant block updates be generated in polynomial time without explicitly performing the full Schur transform?

The parameter count is polynomial for fixed local dimension. The remaining algorithmic issue is efficient block access.

29.4 Dynamic packet graphs

Question 29.4. Can local selection rules guarantee polynomial growth of the reachable packet graph for polynomially long circuits with weak but repeated symmetry breaking?

Bounded excitation gives one exact theorem. Broader results may require locality, energy penalties, or cancellation of packet-transfer paths.

29.5 Minimal packet complexity

Question 29.5. Given a circuit and error tolerance ε, what is the minimum active representation size among all efficiently applicable exact or approximate packet decompositions?

This optimization should include internal encoding and transfer sparsity, not only projector rank.

29.6 Adaptive leakage

Question 29.6. Can one prove polynomial active packet growth for physically local Hamiltonians with weak symmetry breaking over polynomial time?

The commutator bound is local in time and often conservative. Stronger bounds may use energy gaps, locality, Lieb–Robinson estimates, or destructive interference among leakage channels.

29.7 Permutation and Schur compilers

Question 29.7. Which noncollective but globally permutation-invariant circuit layers admit sparse Schur-basis updates without constructing the full Schur transform?

29.8 Orbit quotients under succinct access

Question 29.8. For which succinct graph families are orbit counts, orbit sizes, and quotient couplings computable in polynomial time without enumerating the graph?

29.9 Hybrid Gaussian and packet methods

Question 29.9. Can a polynomial number of non-Gaussian packet corrections be maintained on top of a Gaussian reference for broad interacting few-particle dynamics?

29.10 Open systems

Question 29.10. When does the number of reachable population and coherence blocks remain polynomial under a local Lindblad generator?

29.11 Universal packetization

Question 29.11. Is every polynomial-size quantum circuit classically compressible by some efficiently discoverable representation of polynomial size?

A positive answer would imply $P = BQP$. The record does not support this as a universal empirical principle.

29.12 Interference lower bounds

Question 29.12. Can one prove an explicit representation-independent exponential lower bound for the classical cost of the coherent interference needed by a uniform BQP language?

This is the hard-side frontier. Lower bounds against one simulator family are intermediate results, not the final theorem.

Open program or boundary

The most achievable next theorem is not a universal complexity result. It is a complete compiler theorem for one new symmetry family, with explicit discovery, update, measurement, and leakage algorithms.

Chapter 30

Conclusion

The quantum state vector is a universal coordinate system, not always the computationally intrinsic one. Conserved charges, finite-group representations, permutation orbits, few-particle sectors, and operator packets can expose much smaller dynamics. When the packet is efficiently discoverable, polynomial in size, closed under the gates, and compatible with the measurement, the quantum computation collapses exactly to a deterministic classical polynomial-time simulation.

The strongest current examples are concrete. Fixed charge at constant particle number is polynomial. Exact charge conservation is not necessary: moving packet schedules, efficiently computable moving frames, and bounded unions of low-excitation sectors also collapse to polynomial classical simulation. Fully symmetric fixed-local-dimension dynamics is polynomial. A stronger mixed-state theorem compresses all permutation-invariant Schur sectors into a polynomial commutant record, with $O(n^3)$ density-matrix parameters for qubits under efficient block access. Polynomial-orbit walks are polynomial under quotient-access hypotheses. Few-particle bosonic and fermionic systems are polynomial under efficient updates, while growing particle number gives a sharp progression through quasipolynomial, subexponential, and exponential sector sizes. Stabilizer and Gaussian representations show that an exponential packet can also be easy when its internal states admit compact closed coordinates. Recorded number-conserving benchmarks show large memory and repeated-application gains with exact reconstruction.

The boundaries are equally sharp. Parity, modulo-four charge, and half filling do not generally change exponential scaling. Entanglement can remain hard inside one packet. A syndrome label does not compress logical information. A block encoding does not provide a classical access model. Global Fourier interference may be fully structured yet resist small packetization. Symmetry collapse is therefore not a slogan that all quantum computation is classical. It is a theorem architecture for identifying substantial subclasses that are.

The core research invariant is the complete classical representation record

$$(\text{discovery cost}, N_{\text{packets}}, D_{\text{internal}}, S_{\text{transfer}}, L_{\text{leakage}}, \text{measurement cost}).$$

A future compiler that computes and certifies this record can decide when the full state vector is unnecessary, when a hybrid representation is preferable, and when symmetry alone is insufficient.

Appendix A

Proof details and auxiliary derivations

A.1 Residue projector formula

Let $Q|\psi_k\rangle = k|\psi_k\rangle$. Then

$$\frac{1}{M}\sum_{\ell=0}^{M-1}\omega_M^{-r\ell}e^{2\pi i\ell Q/M}|\psi_k\rangle = \left(\frac{1}{M}\sum_{\ell=0}^{M-1}\omega_M^{(k-r)\ell}\right)|\psi_k\rangle.$$

The bracket equals one when $k \equiv r \pmod{M}$ and zero otherwise. This proves Equation (2.1) and the orthogonality of the residue packets.

A.2 Circuit leakage telescoping

Let $V_j = PG_jP$. Insert and subtract one factor at a time:

$$G_m\cdots G_1P - V_m\cdots V_1P = \sum_{k=1}^{m} G_m\cdots G_{k+1}(G_k - V_k)V_{k-1}\cdots V_1P.$$

Every G_j has norm one and every V_j has norm at most one. On $P\mathcal{H}$,

$$(G_k - V_k)P = (I - P)G_kP.$$

Taking norms gives Theorem 5.2.

A.3 Hypercube quotient coupling

Let

$$|k\rangle = \binom{n}{k}^{-1/2}\sum_{|x|=k}|x\rangle.$$

Every weight-k string has $n-k$ neighbors of weight $k+1$, while every weight-$(k+1)$ string has $k+1$ neighbors of weight k. Counting the bipartite edges yields

$$\langle k+1|A|k\rangle = \frac{\binom{n}{k}(n-k)}{\sqrt{\binom{n}{k}\binom{n}{k+1}}} = \sqrt{(k+1)(n-k)},$$

proving Equation (13.1).

A.4 Symmetric-subspace dimension

A symmetric basis state of n qudits is indexed by occupation numbers

$$(n_1, \ldots, n_d), \qquad n_1 + \cdots + n_d = n.$$

The stars-and-bars count is

$$\binom{n+d-1}{d-1},$$

proving Equation (10.1).

A.5 Bosonic and fermionic dimensions

Bosonic occupations are weak compositions of N into m parts, giving $\binom{N+m-1}{N}$. Fermionic occupations are N-element subsets of m modes, giving $\binom{m}{N}$.

A.6 Acceptance-probability stability

If normalized states $|\psi\rangle, |\phi\rangle$ satisfy

$$\||\psi\rangle - |\phi\rangle\| \le \varepsilon$$

and $0 \le M \le I$, then

$$|\langle\psi|M|\psi\rangle - \langle\phi|M|\phi\rangle| \le 2\varepsilon.$$

Indeed, add and subtract $\langle\phi|M|\psi\rangle$ and use Cauchy–Schwarz. This converts state-vector leakage bounds into measurement-error bounds.

A.7 Orbit quotient invariance

For an automorphism group G, the adjacency matrix commutes with the permutation representation. The fixed subspace of the action is spanned by normalized orbit states. Therefore adjacency preserves that span. Polynomial-time simulation additionally requires direct computation of the quotient entries.

Appendix B

Python API and reproduction commands

The release code is intentionally compact. Its purpose is to verify the algebra and provide a reusable seed rather than to compete with optimized quantum simulators.

B.1 Core API

Listing B.1: Core packet API.

```
from symmetry_packets import (
    weight_basis,
    residue_basis,
    symmetric_subspace_dimension,
    boson_fixed_number_dimension,
    fermion_fixed_number_dimension,
    PacketPlan,
    rz,
    xy_exchange,
    hypercube_quotient,
    circuit_leakage_bound,
    hamiltonian_leakage_bound,
)
```

B.2 Exact packet plan

Listing B.2: Fixed-weight packet compilation.

```
plan = PacketPlan.fixed_weight(n=10, k=2)
packet_state = random_packet_state(plan, seed=12)
full_state = plan.embed(packet_state)

restricted_gate = plan.restrict_gate(
    xy_exchange(theta=0.2),
    qubits=(3, 4),
)
packet_state = restricted_gate @ packet_state
```

B.3 Validation

From the project root, run

```
PYTHONPATH=code python3 code/validate.py
```

The expected output reports successful formula checks and reconstruction errors near machine precision for the representative fixed-weight circuits.

B.4 Figure generation

All figures in the PDF are generated by

```
python3 code/make_figures.py
```

The generator writes vector PDF figures into the `figures` directory.

B.5 Build

The complete rebuild command is

```
./run_all.sh
```

It validates the Python code, regenerates figures, compiles the LaTeX source, and updates the checksums.

Appendix C

Benchmark data ledger

C.1 Exact weight two

The JSON file `data/benchmark_records.json` contains the values in Table 8.1, together with the parity, half-filled, and leakage records.

C.2 Half filling

Table C.1: Recorded half-filled benchmark.

n	k	full dim.	sector dim.	apply gain	one-shot gain	memory gain
8	4	256	70	1.212	1.083	3.657
10	5	1,024	252	1.234	0.7513	4.063
12	6	4,096	924	2.083	0.5092	4.433
14	7	16,384	3,432	3.24	0.2092	4.774
16	8	65,536	12,870	7.041	1.501	5.092
18	9	262,144	48,620	7.757	0.7458	5.392
20	10	1,048,576	184,756	7.396	0.6529	5.675

All recorded relative errors were zero at the displayed precision. The one-shot result can be below one because compilation dominates a single application.

C.3 Parity

Table C.2: Recorded even-parity benchmark.

n	full dim.	packet dim.	apply gain	one-shot gain	memory gain
8	256	128	1.06	1.19	2.0
10	1,024	512	1.15	1.33	2.0
12	4,096	2,048	1.49	1.37	2.0
14	16,384	8,192	1.78	1.75	2.0
16	65,536	32,768	2.0	3.7	2.0

C.4 Symmetry-breaking controls

Test	retained probability	leakage
Break exact weight with pair gate	0.891	0.109
Break parity with X rotation	0.869	0.131
Break weight modulo four with pair gate	0.928	0.0723

These controls demonstrate that a packet restriction must be audited after every symmetry-breaking operation.

Appendix D

Theorem and evidence-status ledger

Table D.1: Claim-status ledger.

Statement	Status	Qualification
Invariant-basis simulation	Proved	Exact gatewise subspace invariance.
Polynomial packet simulation	Proved	Requires efficient discovery, updates, measurement, and polynomial representation size.
Fixed k number-preserving circuits lie in P	Proved corollary	k fixed and local restricted gates efficiently applicable.
Approximate circuit leakage bounded by sum of gate leakages	Proved	Operator-norm state-error certificate.
Hamiltonian leakage bounded by $\|t\|\|\|[H, P]\|\|$	Proved	Bounded self-adjoint Hamiltonian.
Permutation-symmetric qubits lie in P	Derived theorem	Gate matrices and measurement efficient in Dicke coordinates.
Fixed-d symmetric qudits lie in P	Derived theorem	Fixed local dimension and efficient occupation updates.
Polynomial-orbit quantum walks lie in P	Derived theorem	Quotient data computable from succinct input.
Fixed-particle boson and fermion circuits lie in P	Derived theorem	Particle number fixed and updates efficient.
Compressed-packet theorem	Proved abstraction	Representation class closed under gates with polynomial size.
Weight-two runtime and memory gains	Recorded computation	Implementation-specific archival benchmark.
Half-filling remains exponential	Proved dimension statement	Runtime constants remain implementation dependent.
Parity gives only a factor-two dimension reduction	Proved dimension statement	Does not preclude further internal compression.

Statement	Status	Qualification
Quantum Latin order-six product realization absent, entangled realization present	External theorem	Contextual existence defect, not a simulation theorem.
Universal polynomial packetization	Open conjecture	Would imply $P = BQP$.
Representation-independent interference lower bound	Open	Required for a full hard-side complexity conclusion.

D.1 Publication discipline

Every extracted theorem paper should declare the circuit model, input encoding, gate precision, measurement, symmetry discovery method, packet representation, error norm, and complexity parameter. Benchmarks should be accompanied by code and frozen data. Open conjectures should not be phrased as consequences of finite experiments.

Appendix E

Glossary

Term	Meaning
Active packet	A symmetry sector or representation component reached by the input and gates.
Approximate packet	A subspace or representation that is not exactly invariant but has certified leakage.
Compressed packet	A packet whose states are represented by a compact classical data structure rather than explicit amplitudes.
Coherence block	$P_\alpha \rho P_\beta$ with possibly distinct left and right packet labels.
Equitable partition	A graph partition whose cells have uniform neighbor counts, yielding an invariant quotient subspace.
Global interference	Dependence of the output on coherent phase relations among many contributions or packets.
Invariant packet	A subspace mapped into itself by every gate in the declared circuit decomposition.
Leakage	Amplitude or probability transferred outside a proposed active representation.
Multiplicity space	Repeated-copy factor accompanying an irreducible representation in a group decomposition.
Orbit state	Normalized uniform superposition over one group orbit of basis states or graph vertices.
Packet transfer graph	Directed graph recording nonzero blocks $P_\alpha G P_\beta$.
Symmetry collapse	Exact or controlled replacement of a quantum evolution by a polynomial classical representation using symmetry or related compression.
Uniform circuit family	Circuits generated from the input by a deterministic polynomial-time procedure.

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